

The Log-Quantile-Density Model of the Volatility Smile:

Validity, Information, and Convention

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Abstract

A volatility smile is a probability distribution read through the Black formula, and a smile fitter must manage three separable concerns: that its output is a genuine distribution (validity), that the fitted curve reflects what the quotes actually determine (information), and that everything the quotes do not determine is chosen openly (convention). Most parametrizations entangle the three. This paper develops a representation that separates them. The model writes the log-forward return as a monotone transport of a logistic variable, with the logarithm of the transport speed a polynomial in the rank; equivalently, it models the log quantile density as two universal boundary terms plus a Legendre expansion. Validity becomes structural: every point of the open parameter set prices from a positive density with unit mean, the single arbitrage condition being one inequality, $\lambda_+ < 1$, which a logistic reparametrization removes entirely. The two endpoint values of the expansion are the exponential tail scales; they fix the exact strip of finite moments and, through Lee’s moment formula, the asymptotic wing slopes in closed form. One computational object carries the model: the upper-share ledger $G(z) = \int_z^\infty e^{x(t)} \rho(t) dt$. Pricing is two ledger entries; the parameter Jacobian follows from an exact envelope-cancellation identity on the same array; the risk-neutral density is one division; the variance-swap strike is the first moment of the same build; and absence of calendar arbitrage between expiries is the pointwise ordering of two ledgers, proved both ways by conjugate duality. The paper is equally explicit about what is not claimed: the coordinates are unconstrained for a flexible exponential-tail class, not for all smiles; fitted tail parameters are priors selected jointly by basis, ridge, and endpoint coupling, not observations; calendar enforcement is a control, not a theorem; and the continuous theorems are earned by the discretization only through explicit certificates and a randomized audit whose worst violations are measured, not assumed. All market examples are fitted from one frozen snapshot of listed equity options, and every reported number regenerates from that file.

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1 Introduction

The problem treated here is easily stated. A listed option market quotes, for one underlying asset and one expiry, the prices of calls and puts at a few dozen strikes. A fitter must produce from these quotes a complete curve — an implied volatility at every strike — that reproduces the quotes to within their own uncertainty, extends beyond them sensibly, and can be differentiated, integrated, and compared across expiries by every system downstream. The problem looks like curve fitting. It is not, and the difference is the subject of this paper.

Consider the canonical failure. A fitter parametrizes implied volatility $\sigma(k)$ by some pleasant smooth curve, fits seven liquid strikes, and reports residuals of a fraction of a basis point. A risk system then differentiates the fitted call prices twice with respect to strike — because that second derivative *is* the risk-neutral probability density, as section 2 recalls — and finds it negative between two quoted strikes. The fitted surface now prices a butterfly spread — a static portfolio of three vanilla options — at a negative premium: the fitter is offering free money, and reporting a negative probability. Nothing failed numerically. The parametrization simply did not know that a smile is not free to be an arbitrary curve: it must be a probability distribution seen through the Black formula, and the set of such curves has a boundary that a generic parametric family crosses without noticing.

The standard remedy is to keep the pleasant curve and police the boundary: evaluate a density proxy on a grid of strikes, penalize negativity, and hope the optimizer terminates inside the feasible set. This works, mostly, and costs steadily. The constraint is global and nonlinear in the parameters, so it is re-checked at every strike and every iterate; the optimizer spends its effort crawling along the boundary, where penalty gradients fight data gradients; and in sparse or stressed markets — precisely when a fitter is most needed — the boundary is where the solution wants to live.

This paper starts from a different question: *in which representation is validity free?* That is, for which coordinates on the space of smiles is every coordinate vector automatically arbitrage-free, so that nothing is policed and the optimizer roams an open set? The answer developed here is the *log-quantile-density* (LQD) model: write the log-forward return as a monotone transport of a fixed logistic random variable, and parametrize the *logarithm of the transport speed* by a polynomial in the rank. Positivity of the density is then structural — an exponential is positive — the unit-mean condition is absorbed by one scalar shift, and the single remaining condition, one inequality on one number, is removed by a final change of variable. The precise scope should be stated at once, because it is the paper’s first honesty obligation: the construction furnishes *unconstrained coordinates for a flexible exponential-tail class of distributions*, not coordinates for the space of all smiles. Atoms, bounded support, density gaps, and Gaussian or faster tails lie outside the family; section 5 says exactly what is covered and what is approximated.

1.1 Three concerns, separated

The design principle organizing the paper is that a smile fitter must manage three separable concerns, and that the representation should separate them.

1. **Validity** — no static arbitrage. In the LQD representation this is a property of the coordinates, not of the optimizer: every point of the open parameter set is a genuine probability measure with unit mean (section 5). There is no positivity penalty anywhere in the construction, because there is nothing to penalize.
2. **Information** — what the quoted strip actually determines. The body of the distribution is pinned by quotes; the far tails are selected by the basis, the regularization, and the endpoint coupling. The paper says this out loud and quantifies it: the moment strip and

Lee slopes are exact functions of two fitted numbers (section 6), but those numbers are priors dressed as parameters, and the honest deliverable at a traded strike is the effective wing slope, which differs from the asymptotic limit by economically large factors — in either direction, on one and the same fitted slice (fig. 7).

3. **Convention** — what the fitter chooses where the data is silent. Basis order, ridge weight, the optimization chart, the scope of calendar enforcement: each is an explicit, auditable choice, orthogonal to validity (sections 9, 11 and 12). A convention defended as a theorem is a future incident; the paper labels its conventions as such.

One computational object carries all of this, and it plays the role in this paper that a single unifying quantity plays in any theory worth the name. Define the *upper-share ledger*

$$G(z) = \int_z^\infty e^{x(t)} \rho(t) dt,$$

the share of the (unit) forward carried by outcomes above the z -th log-odds level; here x is the transport map and ρ the logistic density, both defined in section 4. Five computations that are usually five pieces of software are five uses of this one array:

- *pricing*: a call is the difference of two ledger entries (section 7);
- *differentiation*: the parameter Jacobian of every price follows from an exact cancellation — the strike root’s motion drops out — so one quadrature pass differentiates the whole fit (section 10);
- *density*: the Breeden–Litzenberger density is one division of arrays already on the grid (section 7);
- *the variance swap*: the log-contract strike is the first moment of the same build (section 7);
- *calendar order*: two expiries are free of calendar arbitrage precisely when their ledgers are pointwise ordered — a theorem proved both ways by conjugate duality (section 11).

1.2 Contributions and route

The ingredients are classical: Breeden and Litzenberger’s density identity [4], Lee’s moment formula [16], quantile-function modeling in the statistical tradition of Parzen [18] and Gilchrist [12], the log-quantile-density transform studied by Petersen and Müller [19], and the log contract [17]. The contribution claimed is the *separation of concerns* — validity structural, information audited, convention explicit — and the *single-ledger computation* that makes the separation cheap enough for production use. Along the way the paper proves the statements a referee would otherwise have to take on faith: the finite-forward wall (proposition 5.2), butterfly-freedom (proposition 5.1), a universality proposition with its exclusions stated (proposition 5.3), closed-form tail-scale-to-Lee-slope maps (proposition 6.3), exact at-the-money level, skew, and curvature with the skew exhibited as a digital mismatch (proposition 8.1), the envelope-cancellation theorem behind the analytic Jacobian (theorem 10.1), and the equivalence of calendar order with ledger order (theorem 11.3).

The paper is written for a reader who knows the Black–Scholes formula and nothing else about smile modeling. Section 2 assembles the constraint set; section 3 builds the representation from a heuristic about ranks and returns; sections 4 and 5 define the family and prove validity; section 6 treats tails and moments; sections 7 and 8 develop the ledger and the at-the-money microscope; sections 9 to 11 cover calibration, differentiation, and the calendar; section 12 gives the numerical analysis, including the certificates by which the discretization earns the

continuous theorems; section 13 works real-market and synthetic examples from one frozen data snapshot; sections 14 and 15 position the work and state its limits. Implementation detail is confined to appendix A; deferred proofs to appendix B; data provenance to appendix C.

2 The constraint set

2.1 Normalized units

Fix one underlying and one expiry T . Everything deterministic — discounting, forwards, carry — divides out before modeling begins. Let F be the forward price to T and D the discount factor; we work under the T -forward measure with the *log-forward return* and *gross return*

$$X = \log \frac{S_T}{F}, \quad Y = e^X = \frac{S_T}{F}. \quad (1)$$

Strikes are quoted in log-moneyness $k = \log(K/F)$, and the *normalized call* is

$$c(k) = \mathbb{E}[(e^X - e^k)^+], \quad \mathbb{E}[e^X] = 1. \quad (2)$$

The second equality is the martingale (forward) condition: the forward is the mean of the terminal spot under its own measure. A dollar quote $c^\$$ converts by $c = c^\$/(DF)$, and nothing further about rates or carry appears in the paper; deterministic carry is a standing assumption, revisited only when expiries are coupled in section 11. The normalized put is $P(k) = \mathbb{E}[(e^k - e^X)^+]$; put–call parity is $c(k) - P(k) = 1 - e^k$.

One symbol of time suffices, but it must be the right one. Desks distinguish the expiry date, its calendar year fraction, and the *variance time* — calendar time possibly dilated around scheduled events so that implied variance accrues at the market’s rate rather than the clock’s. Everything in this paper that multiplies a variance, scales a vega, or annualizes a number uses the variance time, written τ ; when no event dilation is in force, τ is the calendar fraction, and nothing below depends on which convention feeds it. Total implied variance at log-moneyness k is $w(k) = \sigma^2(k) \tau$, with σ the Black implied volatility defined through the normalized Black call

$$B(k, w) = \Phi(d_+) - e^k \Phi(d_-), \quad d_\pm = -\frac{k}{\sqrt{w}} \pm \frac{\sqrt{w}}{2}, \quad (3)$$

by the root $B(k, w(k)) = c(k)$ [2]. A *volatility basis point* (vol bp) is 10^{-4} in volatility.

2.2 Statics: what a slice must satisfy

The reason a smile is not free to be an arbitrary curve is one integration by parts away. Write the call as a function of the normalized strike $y = e^k$. When Y has a density f_Y ,

$$\frac{\partial c}{\partial y} = -\mathbb{P}(Y > y), \quad \frac{\partial^2 c}{\partial y^2} = f_Y(y) \geq 0. \quad (4)$$

This is the Breeden–Litzenberger identity [4]: the call curve determines the distribution, and every property of the call curve that a static portfolio can monetize is a property of the distribution. A *butterfly* of half-width δ — long one call at $y - \delta$, long one at $y + \delta$, short two at y — costs $c(y - \delta) - 2c(y) + c(y + \delta)$, and by (4) its price per δ^2 converges to the density at y . A negative butterfly price is a negative density, exactly.

Definition 2.1 (Arbitrage-free slice). A *slice* is a call curve $c(k)$ of the form (2) for some law of X with $\mathbb{E}[e^X] = 1$. Equivalently, for laws with a density on $(0, \infty)$: as a function of the strike $y = e^k$, c is decreasing and convex, $(1 - y)^+ \leq c(y) \leq 1$, the slope lies in $[-1, 0]$ with $c'(y) \rightarrow -1$ as $y \rightarrow 0^+$ (anything shallower at zero strike would put mass at $S_T = 0$), and $c(y) \rightarrow 0$ as $y \rightarrow \infty$ [6].

The space of slices is therefore, up to these boundary conditions, the space of mean-one probability laws on $(0, \infty)$. That is the object a fitter must parametrize. Note one caveat now, because it recurs: convexity holds in the strike y , *not* in log-moneyness k . In k the chain rule inserts factors of e^k , and a deep in-the-money call is legitimately concave in k . Audits that check convexity must check it in the right variable (section 12).

2.3 Why these constraints are awkward in volatility space

Every trader quotes $\sigma(k)$, so the temptation is to parametrize it directly; SVI [10, 11] is the outstanding example. The cost appears when (4) is translated into the volatility chart. Substituting $c = B(k, w(k))$ and differentiating twice through the Black formula, the density of X factors exactly as

$$f_X(k) = g_D(k) \frac{\varphi(d_-(k, w(k)))}{\sqrt{w(k)}}, \quad g_D(k) = \left(1 - \frac{k w'(k)}{2w(k)}\right)^2 - \frac{w'(k)^2}{4} \left(\frac{1}{w(k)} + \frac{1}{4}\right) + \frac{w''(k)}{2}, \quad (5)$$

with d_- as in (3): no-arbitrage in the volatility chart is $g_D(k) \geq 0$ for all k , the condition studied by Gatheral and Jacquier [11, §2]. The factorization fixes the normalization of g_D , and two checkpoints let a reader verify it without redoing the two-page differentiation. For a flat smile ($w' = w'' = 0$), $g_D \equiv 1$ and (5) reduces to the Black density of X exactly. And at the money, substituting the exact derivatives of section 8 gives $g_D(0) = \sqrt{w_0} f_X(0)/\varphi(d) \geq 0$ — the identity (5) read backwards at one point. Three features make the condition $g_D \geq 0$ expensive to live with. It is *global*: a violation can appear at any k , including between quotes, so it must be monitored on a grid. It is *nonlinear* in the parameters, entering through w , w' , and w'' simultaneously, so no useful parametric family satisfies it identically. And it is *coupled*: the martingale condition and the wing growth bounds (section 6) constrain the same parameters from other directions at the same time. The volatility chart is the most convenient to quote and the most expensive to police. The representation problem of this paper is to find the chart where the policing cost is zero, and the next section builds it from first principles.

2.4 The notation ledger

The paper insists on few symbols. Table 1 lists them all; no ledger symbol is reused with a second meaning (bound variables local to a single derivation, such as the butterfly half-width δ above or the substitutions inside one proof, stay local and never enter the ledger).

3 Heuristic: from ranks to returns

The construction rests on one elementary change of viewpoint, and it is worth presenting slowly, because everything after it is bookkeeping.

3.1 A distribution is a rule assigning returns to ranks

A probability law for the return X is usually pictured as a density: how much probability sits near each return level. Invert the picture. Order all possible outcomes from worst to best and index them by their *rank* $u \in (0, 1)$: $u = 0.01$ is the first percentile, $u = 0.5$ the median, $u = 0.99$ the ninety-ninth. The law is then completely described by the rule that assigns to each rank the return realized there,

$$u \longmapsto Q(u), \quad (6)$$

the *quantile function*, the inverse of the CDF. If U is uniform on $(0, 1)$ then $X = Q(U)$ has exactly the law described — this is the inverse-CDF construction used by every random-number

$X, Y = e^X$	log-forward return; gross return	$u, z = \text{logit } u$	rank; log-odds
$k; c(k), P(k)$	log-moneyness; norm. call, put	Λ, ρ	logistic CDF, density
$Q, q = Q'$	quantile of X ; quantile density	g	log transport speed (in rank)
$x(z), m$	transport map; martingale shift	G	upper-share ledger
w, σ, τ	total variance $\sigma^2\tau$; vol; variance time	$\theta = (L, R, a)$	coefficients; P_n Legendre
λ_-, λ_+	endpoint speeds $e^{g(0)}, e^{g(1)}$	β, Ψ, r	Lee slope; Lee map; moment order
$B(k, w)$	normalized Black call (3)	$d = \sqrt{w_0}/2$	ATM Black argument
u_k, z_k	rank and log-odds of strike k	u_*, f_*, Δ	forward rank; ATM density; digital gap
α, ν	ridge weight, ridge power	η, h	vega floor; haircut half-width
ξ	logistic-chart coordinate, $\lambda_+ = \text{expit } \xi$	$N, Z_{\max}$	basis order; quadrature half-width

Table 1: Every recurring symbol in the paper. Φ and φ are the standard normal CDF and PDF; $\text{expit}(z) = 1/(1 + e^{-z})$ and logit is its inverse. One differentiation convention holds throughout: a *prime* denotes the derivative of a function of one variable (c', w', σ', G'); a *subscripted* ∂ denotes a partial derivative of a function of several variables ($\partial_k B, \partial c/\partial \theta_j$). Z is the logistic score variable of section 4; $Z_{\max}$, and nothing else, is the quadrature half-width in log-odds. g_D (5) is the volatility-chart density factor, distinct from the model’s log-speed g .

generator. Two facts make the rank picture attractive for our problem. First, the horizontal axis is *universal*: every law, however wild, is a rule on the same interval $(0, 1)$, and the uniform weight on ranks is the same for all of them — ranks are born normalized, so “total probability one” is not a constraint but a tautology. Second, the entire content of “being a probability law on returns” collapses to a single property: Q must be *increasing*. Nothing else.

Monotonicity is a far better constraint to own than positivity of a second derivative, because it is destroyed by one differentiation, not two. Let

$$q(u) = Q'(u) \quad (7)$$

be the *quantile density*: the spacing of returns per unit of rank. Then Q increasing $\iff q > 0 \iff \log q$ is an arbitrary real function. Modeling $\log q$ therefore removes the last pointwise constraint. The quantity $\log q$ has a physical reading that will guide every design choice below: it is the *log-speed* at which the return axis is traversed as the rank advances. Where the speed is low, ranks are packed into a narrow band of returns — high probability density; where the speed is high, a few ranks are stretched across a wide band — low density. Indeed, differentiating $F_X(Q(u)) = u$,

$$f_X(Q(u)) = \frac{1}{q(u)} : \quad (8)$$

density is the reciprocal of speed. A model of $\log q$ is a model of the distribution in which positivity of the density can never fail, for the same reason e^t can never be negative.

3.2 The two singularities, and the coordinate that removes both

The speed q cannot be an arbitrary *bounded* function, because an unbounded law must escape to $\pm\infty$ as $u \rightarrow 0$ or 1 : the integral of q over $(0, 1)$ is the total return range, infinite for any law with unbounded support. So $\log q$ must diverge at both endpoints, and the divergence is not a nuisance to be squashed but a structure to be normalized. Ask how fast the divergence is for the benchmark laws. For a standard Gaussian, $q(u) = 1/\varphi(\Phi^{-1}(u))$, which blows up like $1/(u\sqrt{2\log(1/u)})$ as $u \rightarrow 0$ (using the Gaussian tail equivalent $\Phi(x) \sim \varphi(x)/|x|$ as $x \rightarrow -\infty$, so $\varphi(\Phi^{-1}(u)) \sim u|\Phi^{-1}(u)| \sim u\sqrt{2\log(1/u)}$) — essentially $1/u$, with a slowly varying correction. For an exponential tail, $q(u)$ blows up like a constant times $1/u$ exactly. The pattern is

general: the endpoint divergences of $\log q$ are, at leading order, $-\log u$ at the left end and $-\log(1-u)$ at the right, with the law's individual character riding on a lower-order remainder. For exponential tails the remainder is *bounded* — it tends to the logarithm of the tail scale — while for the Gaussian it still drifts to $-\infty$, slowly. Boundedness of the remainder is precisely what will delimit the model's class (proposition 5.3).

This suggests separating skeleton from flesh,

$$\log q(u) = \underbrace{-\log u - \log(1-u)}_{\text{universal skeleton}} + g(u), \quad (9)$$

and modeling only the smooth part g . There is a cleaner way to say the same thing, and it is the form the whole computational apparatus will use. Change the rank coordinate to *log-odds*,

$$z = \text{logit } u = \log \frac{u}{1-u}, \quad u = \Lambda(z) = \frac{1}{1+e^{-z}}, \quad (10)$$

so that $z = 0$ is the median and $z = \pm \log 9 \approx \pm 2.20$ are the 90th and 10th percentiles: equal steps in z multiply the odds $u/(1-u)$ by a constant factor. The derivative of the map is $du/dz = \Lambda(z)(1-\Lambda(z)) = \rho(z)$, the logistic density, which decays like $e^{-|z|}$ in both directions. Now compute the speed of the return assignment *per unit of log-odds*:

$$\frac{dQ}{dz} = q(u) \frac{du}{dz} = \frac{e^{g(u)}}{u(1-u)} \cdot u(1-u) = e^{g(u)}. \quad (11)$$

Both endpoint singularities cancel identically. In the log-odds coordinate the quantile of any law in our class is the integral of a smooth, strictly positive, *bounded* function — the exponential of the very g we chose to model. The coordinate z does for ranks what the logarithm does for prices: it moves the interesting boundary behavior to a place where it is linear.

3.3 The model in one sentence

Assemble the pieces. Take Z a standard logistic variable (that is, $Z = \text{logit } U$ for U uniform — CDF Λ , density ρ), and define the return by transporting Z through the integrated speed:

$$X = x(Z), \quad x(z) = m + \int_0^z e^{g(\Lambda(t))} dt, \quad (12)$$

with g a polynomial of modest degree in the rank and m the scalar that makes $\mathbb{E}[e^X] = 1$. The map x is strictly increasing whatever the polynomial, so X always has a genuine law; the density is positive by (8); the mass is normalized because ranks are; the mean condition costs one scalar. What the polynomial's two endpoint values control — the asymptotic speeds $e^{g(0)}$ and $e^{g(1)}$ — turns out to be exactly the exponential decay scales of the two return tails, and the single validity condition of the entire construction will be the requirement that the right tail decay fast enough to give e^X a finite mean. Figure 1 draws the three objects for the simplest possible choice, g constant: the logistic score with its percentile marks, the resulting affine transport with the strike roots of two options marked on it, and the flat-at-the-money smile it prices. The next section states the definition precisely and dispatches the constant-speed case in closed form.

4 The LQD family

4.1 Definition

It remains to choose the smooth part g , and the choice is guided by two requirements: the endpoint values must be readable — they will turn out to be the tail scales — and the body

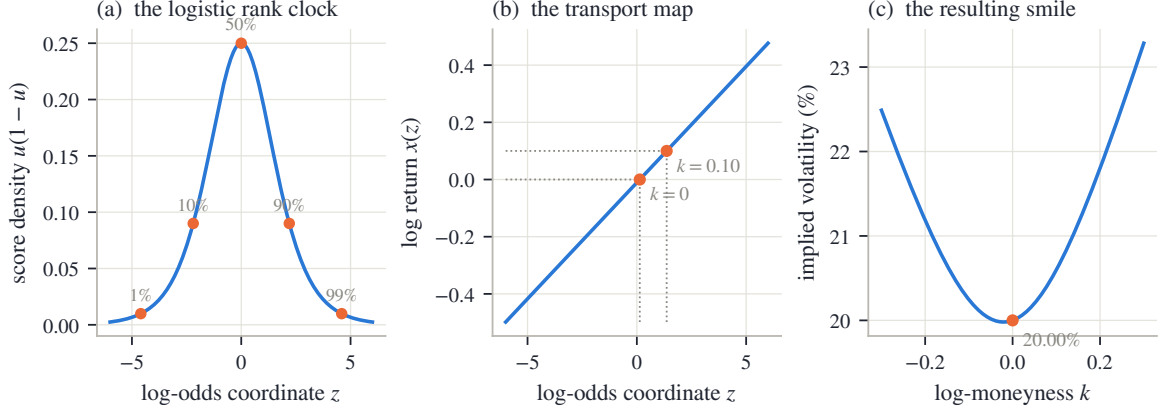


Figure 1: The transport view for a constant-speed slice. Panel A: the logistic score density $\rho(z)$ on the log-odds line; the marked ticks are the 1st, 10th, 50th, 90th, and 99th percentiles — every percentile of an unbounded law at a finite coordinate. Panel B: the transport map $x(z)$, an exact straight line when the log-speed g is constant; the dots mark the strike roots z_k solving $x(z_k) = k$ for the at-the-money strike $k = 0$ and one out-of-the-money strike — the two options priced by hand on this very slice in section 7. Panel C: the implied-volatility smile of this slice — flat at the money and gently convex, because exponential return tails carry more far-strike value than the log-normal’s, which the Black formula can only express as higher implied volatility.

must be flexible, well-conditioned, and cheap. A polynomial in the rank, expressed in the Legendre basis on $u \in [0, 1]$, meets all three. The central definition of the paper is the following; it is the only boxed display.

Definition 4.1 (The LQD slice). Fix an order $N \geq 2$ and a coefficient vector $\theta = (L, R, a_2, \dots, a_N) \in \mathbb{R}^{N+1}$. Let Z be a standard logistic variable, with CDF $\Lambda(z) = 1/(1 + e^{-z})$ and density $\rho(z) = \Lambda(z)\Lambda(-z)$. The LQD slice is the law of

$$X = x(Z), \quad x(z) = m + \int_0^z e^{g(\Lambda(t))} dt, \quad g(u) = (1-u)L + uR + \sum_{n=2}^N a_n P_n(1-2u), \quad (13)$$

where P_n is the Legendre polynomial of degree n and the scalar m is fixed by the martingale condition $\mathbb{E}[e^X] = 1$.

Three remarks unpack the display. First, the map x is strictly increasing for *every* θ , since its derivative is an exponential; so X always has a genuine law, with CDF and density

$$F_X(x(z)) = \Lambda(z), \quad f_X(x(z)) = \frac{\rho(z)}{x'(z)} = \Lambda(z)\Lambda(-z) e^{-g(\Lambda(z))} > 0. \quad (14)$$

This single line is the entire no-arbitrage mechanism of section 5; no positivity condition is ever checked, because none can fail. Second, the quantile function of X is $Q(u) = x(\text{logit } u)$, and by (11) its quantile density is

$$q(u) = \frac{e^{g(u)}}{u(1-u)}, \quad \text{i.e.} \quad \log q(u) = -\log u - \log(1-u) + g(u) : \quad (15)$$

the definition is exactly the skeleton-plus-smooth-part decomposition (9) with a polynomial g — hence the name *log-quantile-density* model. The transport form (13) and the LQD form (15)

are the same object; the paper computes in the first and reasons about distributions in the second. Third, the expansion starts at $n = 2$ deliberately: degrees zero and one are already spanned by the *endpoint chord* $(1 - u)L + uR$, since $(1 - u)L + uR = \frac{L+R}{2} + \frac{L-R}{2}(1 - 2u)$ and $P_0 = 1$, $P_1(1 - 2u) = 1 - 2u$. The family at order N is thus exactly the set of laws whose g is a polynomial of degree $\leq N$; the chord is a choice of *coordinates* on that space, made so that the two leading coefficients sit at the endpoints, where the tails live. Nothing in the definition needs N larger than 2; the reference implementation restricts to $N \geq 4$, a calibration convention aligned with the ridge of section 9 (which starts penalizing at $n = 4$) rather than a mathematical requirement. Legendre itself is also a disclosed convention: any polynomial basis spans the same family, but the Legendre modes are orthogonal under the uniform rank measure — the inner product every rank integral in the model uses — so high-order coordinates remain numerically distinguishable in the least-squares normal equations, where monomials in u would reproduce Hilbert-matrix conditioning. Chebyshev would serve comparably; monomials would not.

Because $P_n(1) = 1$ and $P_n(-1) = (-1)^n$, the two endpoint values of g are

$$g(0) = L + \sum_{n \geq 2} a_n, \quad g(1) = R + \sum_{n \geq 2} (-1)^n a_n, \quad (16)$$

and we name their exponentials the *endpoint speeds*

$$\lambda_- = e^{g(0)}, \quad \lambda_+ = e^{g(1)}. \quad (17)$$

By (11) these are the asymptotic speeds of the transport. Precisely, writing $\bar{x}(z) = \int_0^z e^{g(\Lambda(t))} dt$ for the *unshifted* transport (so $x = m + \bar{x}$), there are constants b_- , b_+ with

$$\bar{x}(z) = b_- + \lambda_- z + O(e^z) \quad (z \rightarrow -\infty), \quad \bar{x}(z) = b_+ + \lambda_+ z + O(e^{-z}) \quad (z \rightarrow +\infty), \quad (18)$$

because g is Lipschitz on $[0, 1]$ and $\Lambda(z)$ approaches its endpoint at the rate $e^{-|z|}$, so the speed $e^{g(\Lambda(z))}$ approaches $\lambda_{\pm}$ at that same rate and the deviation integrates to a convergent constant plus an exponentially small tail. The shifted transport obeys the same asymptotes with constants $m + b_{\pm}$; keeping the two families of constants apart matters below (the tail prefactor of proposition 6.1 contains $m + b_+$, not b_+). This estimate does real work in proposition 4.2 and section 6. Note, for later honesty, that (16) couples body and tails in the raw coordinates: every a_n moves one or both endpoint values. Adding 0.10 to a_2 multiplies *both* speeds by $e^{0.10} \approx 1.105$. The fitted law is coordinate-independent, but the raw coordinates are a poor language for tail priors; section 12 introduces the two reparametrizations — the endpoint chart and the logistic chart — that repair this.

What does one coefficient do? Raising g near a rank u stretches the return spacing there, and stretched spacing is thinned probability (8): a basis mode is a mass-shuffling instruction written in rank coordinates. Even modes $(P_2, P_4, \dots)$ act symmetrically — $a_2 > 0$ stretches both tails and compresses the center, a sharper peak with fatter tails, which the smile shows as extra convexity; odd modes tilt — $a_3 > 0$ slides mass from one side to the other, which the smile shows as skew; higher modes sculpt with finer brushes. Figure 2 shows the three charts of the same three perturbations.

4.2 The martingale shift

The scalar m deserves one short proposition, because it is the only normalization in the model and everything else stands on it. Recall $\bar{x}$, the unshifted transport of (18), with $x = m + \bar{x}$.

Proposition 4.2 (Martingale normalization). *Let $I = \int_{-\infty}^{\infty} e^{\bar{x}(z)} \rho(z) dz$. If $I < \infty$, then $m = -\log I$ is the unique scalar for which $\mathbb{E}[e^X] = 1$, and the shift changes neither the density's shape, nor its tails, nor its positivity. Moreover $I < \infty$ if and only if $\lambda_+ < 1$.*

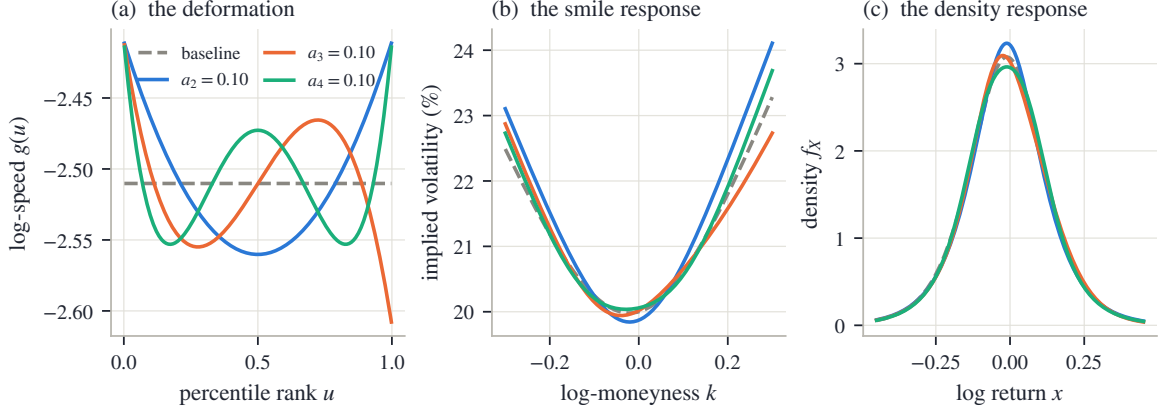


Figure 2: One coefficient at a time: a_2 , a_3 , or a_4 set to 0.10 on a symmetric constant-speed reference, everything else untouched. Panel A: the perturbation of the log-speed g over the rank — even modes symmetric, odd modes not. Panel B: the implied-volatility smile — mode 2 bends it symmetrically, mode 3 tilts it, mode 4 works the shoulders. Panel C: the density — mass thins where g rises and thickens where it falls, and every perturbed density is non-negative without anything being checked. Every raw mode drawn here also moves *both* tail scales, by a factor $e^{\pm 0.10}$: the even modes multiply both by $e^{0.10}$, while a_3 multiplies λ_- by $e^{0.10}$ and λ_+ by $e^{-0.10}$ — $P_n(-1) = (-1)^n$ in (16), visible in panel A as the a_3 curve ending below baseline at $u = 1$. This is the coupling the endpoint chart of section 12 removes.

Proof. Since $X = x(Z)$ and Z has density ρ , $\mathbb{E}[e^X] = \int e^{m+\bar{x}(z)} \rho(z) dz = e^m I$; if I is finite, $e^m I = 1$ has the unique solution $m = -\log I$. The shift translates the law of X by a constant, which multiplies f_X by nothing and moves every quantile equally. For the convergence claim, use (18): as $z \rightarrow +\infty$, $\bar{x}(z) = b_+ + \lambda_+ z + O(e^{-z})$ with b_+ finite, so the integrand is asymptotically e^{b_+} times $e^{(\lambda_+-1)z}$, using $\rho(z) \sim e^{-z}$. This is integrable at $+\infty$ iff $\lambda_+ < 1$; if $\lambda_+ = 1$ the integrand tends to a positive constant, and if $\lambda_+ > 1$ it grows, so the integral diverges in either case. As $z \rightarrow -\infty$, symmetrically, $\bar{x}(z) = b_- + \lambda_- z + O(e^z)$ with $\lambda_- > 0$, so the integrand is asymptotically a constant times $e^{(1+\lambda_-)z}$, integrable at $-\infty$ for every $\lambda_- > 0$: the left end never obstructs. $\square$

The condition $\lambda_+ < 1$ is the *finite-forward wall*, and it is the single validity constraint of the whole construction; section 5 discusses its meaning and section 6 shows it is not an artifact of the basis but a fact about exponential tails.

4.3 An exactly solvable slice

Set every body coefficient to zero and the two chord coefficients equal: $a_n = 0$, $L = R = \log s$ with $0 < s < 1$. Then $g \equiv \log s$, the transport is exactly affine, $x(z) = m + sz$, and X is a logistic random variable with scale s . This slice is to the LQD family what the harmonic oscillator is to mechanics: every formula of the paper can be evaluated on it in closed form, and section 13 uses it as a machine-precision audit of the implementation.

Everything flows from one integral. The moment generating function of the standard logistic score is a rank integral that Euler solved: substituting $u = \Lambda(z)$, $e^{tz} = \left(\frac{u}{1-u}\right)^t$,

$$\mathbb{E}[e^{tZ}] = \int_0^1 \left(\frac{u}{1-u}\right)^t du = B(1+t, 1-t) = \Gamma(1+t)\Gamma(1-t) = \frac{\pi t}{\sin(\pi t)}, \quad |t| < 1, \quad (19)$$

where B is the beta function, the middle equality is its Gamma representation, and the last is Euler's reflection formula. Note that (19) already confirms the wall: the MGF exists exactly

on the strip $|t| < 1$, as section 6 says it must for a law with unit endpoint speeds. Three consequences:

1. *The shift in closed form.* $I = \mathbb{E}[e^{sZ}] = \pi s / \sin(\pi s)$, so $m = -\log(\pi s / \sin(\pi s))$ — the only slice in the family whose normalization needs no quadrature, and therefore the natural end-to-end test vector.
2. *The variance, hence the initializer.* Expanding $\log \mathbb{E}[e^{tZ}] = \frac{\pi^2 t^2}{6} + O(t^4)$ (from $\sin \pi t = \pi t(1 - \pi^2 t^2/6 + \dots)$), the second cumulant is $\text{Var}(Z) = \pi^2/3$, so $\text{Var}(X) = \pi^2 s^2/3$. Matching a target ATM total variance w_0 gives the cold-start seed

$$s = \frac{\sqrt{3w_0}}{\pi}, \quad (20)$$

which is where every calibration in this paper starts when it has no better information (section 9).

3. *The seed's known bias.* Matching the *variance* does not match the *ATM price*. For small s , the at-the-money call of this slice is $c(0) = \mathbb{E}[(e^X - 1)^+] \approx s \mathbb{E}[Z^+] = s \log 2$ (two small steps deserve their receipts: the shift is negligible at this order, $m = -\log \frac{\pi s}{\sin \pi s} = -\frac{\pi^2 s^2}{6} + O(s^4)$, so $(e^X - 1)^+ \approx X^+ \approx sZ^+$; and $\mathbb{E}[Z^+] = \int_0^\infty z \rho(z) dz = \sum_{n \geq 1} \frac{(-1)^{n-1}}{n} = \log 2$, by expanding $\rho(z) = e^{-z}(1 + e^{-z})^{-2}$ in a geometric series and integrating term by term), while the Black price at total variance w is $\approx \sqrt{w/2\pi}$; equating gives an implied ATM total variance $w \approx 2\pi(\log 2)^2 s^2$, and the ratio to the variance-matched target is

$$\frac{2\pi(\log 2)^2}{\pi^2/3} = \frac{6(\log 2)^2}{\pi} \approx 0.918.$$

The cold start therefore under-prices ATM variance by about 8% — a derived constant, confirmed in panel B of fig. 3 — which the first optimizer iteration removes. It is recorded here because a seed whose bias is known is a seed that can be trusted.

Admissibility of the slice is $\lambda_- = \lambda_+ = s < 1$; a 55% volatility at one year is $s \approx 0.30$, nowhere near the wall. Figure 3 confronts the reference implementation with the closed forms across scales $s \in [0.05, 0.95]$: the transport map agrees with $m + sz$ to 1.1×10^{-11} (cumulative-Simpson roundoff), the shift with $-\log(\pi s / \sin \pi s)$ to 3.0×10^{-9} , and the martingale check $|\mathbb{E}[e^X] - 1|$ stays below 3.3×10^{-16} . The three levels are worth distinguishing rather than blurring into “machine precision”: the shift error is the largest and *grows toward the wall*, as it must — the normalizing integrand decays like $e^{(\lambda_+ - 1)z}$, so as $s \rightarrow 1$ its tail thickens and the truncation-corrected quadrature works harder. A wrong grid constant, a mis-anchored cumulative sum, or a stale cache moves any of these agreements by many orders of magnitude, which is what makes the audit sharp.

4.4 A fitted slice in the model's own chart

Before proving anything, it is worth seeing what a real fitted g looks like. Figure 4 takes one liquid equity-index expiry from the frozen snapshot of section 13 (SPY, December 2026; provenance in appendix C) and draws the fitted $\log q(u)$: the universal skeleton $-\log u - \log(1 - u)$ dashed, the fitted smooth part g shaded between the curves. The entire freedom of the model is that shaded band — low-degree, gently sloped, its two endpoint values the fitted tail scales — and *any* smooth band whatever, shifted, bent, or oscillating, is another arbitrage-free slice provided only that the right-hand endpoint stays below zero. The companion panel shows the same object as a density.

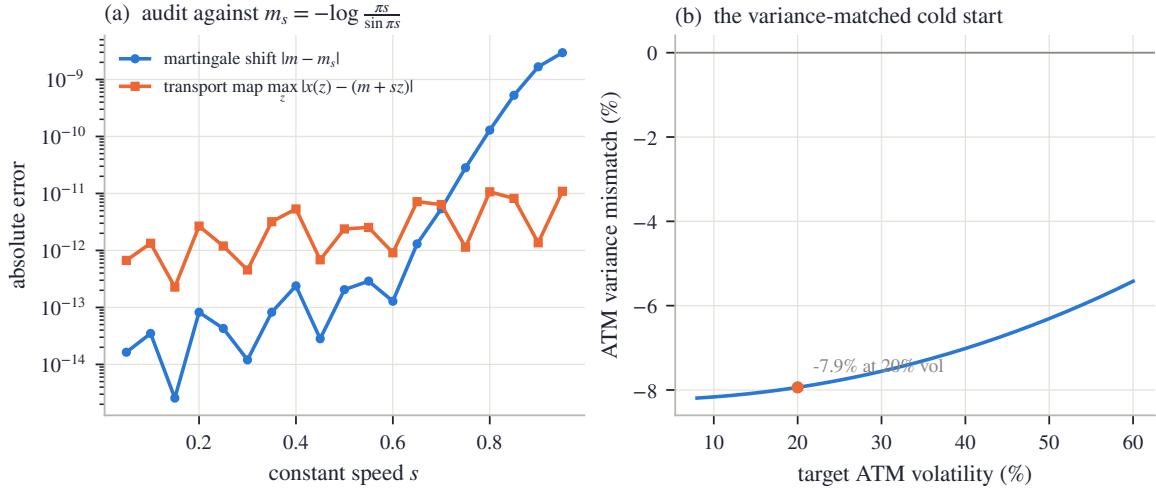


Figure 3: The constant-speed audit, in two panels. Panel A: the implementation's martingale shift m and transport map $x(z)$ against the closed forms $-\log(\pi s / \sin \pi s)$ and $m + sz$ from (19), absolute error on a log scale across the admissible scale range: map error at most 1.1×10^{-11} ; shift error at most 3.0×10^{-9} , climbing as $s \rightarrow 1$ because the normalizing integrand's decay rate $1 - \lambda_+$ vanishes at the wall. Panel B: the cold-start bias of (20) — the ATM variance mismatch of the variance-matched seed, measured -7.9% at the 20%-vol slice, against the derived small- s limit $6(\log 2)^2 / \pi - 1 \approx -8.2\%$; the first calibration step removes it.

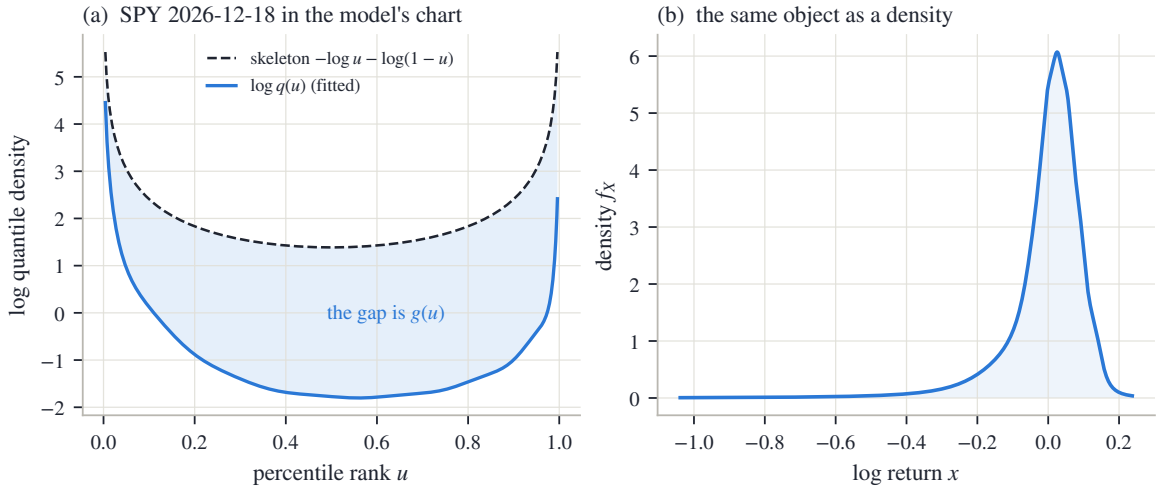


Figure 4: A real fitted slice (SPY, December 2026 expiry, from the frozen snapshot of appendix C) in the model's own chart. Panel A: fitted $\log q(u)$ (solid) over the universal skeleton $-\log u - \log(1 - u)$ (dashed); the shaded gap between them is the fitted g — the model's entire freedom. Its left endpoint sits higher than its right: a fatter left tail, the index put skew in the model's native coordinates. Panel B: the same slice as a return density, skewed left, with the strikes of the quoted strip marked — the region where the data, rather than the basis, chose the curve.

5 Validity theorems

This section proves that validity is structural: every admissible coefficient vector prices from a genuine mean-one law, so the optimizer of section 9 never sees a constraint. “Admissible” means exactly one thing:

$$\Theta_N = \{\theta = (L, R, a_2, \dots, a_N) \in \mathbb{R}^{N+1} : \lambda_+ < 1\}, \quad (21)$$

an *open* half-space-like subset of $\mathbb{R}^{N+1}$ cut out by one smooth scalar inequality — and section 12 shows a change of variable under which even that inequality disappears, the admissible set becoming all of $\mathbb{R}^{N+1}$.

5.1 One expiry: butterfly-freedom without a certificate

Proposition 5.1 (One-expiry no-arbitrage). *Let $\theta \in \Theta_N$ and let $c(k) = \mathbb{E}[(e^X - e^k)^+]$ be priced from the slice (13). Then:*

- (i) *X has a strictly positive, continuous density on $\mathbb{R}$, and $Y = e^X$ a strictly positive density on $(0, \infty)$ with $\mathbb{E}[Y] = 1$;*
- (ii) *as a function of the strike $y = e^k$, the call is strictly decreasing and strictly convex, with $(1 - y)^+ \leq c \leq 1$, slope $c'(y) = -\mathbb{P}(Y > y) \in (-1, 0)$, and $c \rightarrow 0$ as $y \rightarrow \infty$;*
- (iii) *every finite-width butterfly has strictly positive price, and put–call parity $c(k) - P(k) = 1 - e^k$ holds identically;*
- (iv) *convexity in log-strike fails where it should: c is concave in k on a deep in-the-money region, and this is a property of every arbitrage-free slice, not a defect.*

Proof. (i) is (14) together with proposition 4.2: x is a C^1 diffeomorphism of $\mathbb{R}$ (its derivative $e^{g(\Lambda(z))}$ is continuous and positive), so $X = x(Z)$ has density $\rho(x^{-1}(\cdot))/x'(x^{-1}(\cdot)) > 0$; the shift m makes $\mathbb{E}[e^X] = 1$, and $Y = e^X$ inherits a positive density on $(0, \infty)$ by another smooth change of variable.

(ii) Write the call rank-by-rank: with U uniform and $Y = e^{Q(\text{logit } U)}$ — hereafter we write $Q(u)$ for $x(\text{logit } u)$ —

$$c(y) = \int_0^1 (e^{Q(u)} - y)^+ du.$$

For fixed u the integrand is convex and non-increasing in y ; both properties survive the integral. Strictness and the slope value follow from differentiating under the integral (dominated by 1): $c'(y) = -\int_0^1 \mathbf{1}\{e^{Q(u)} > y\} du = -\mathbb{P}(Y > y)$, which lies strictly between -1 and 0 because the density of Y is positive everywhere, and is strictly increasing in y for the same reason, which is strict convexity. The bounds are $\mathbb{E}[(Y - y)^+] \leq \mathbb{E}[Y] = 1$ and $(1 - y)^+ = (\mathbb{E}[Y] - y)^+ \leq \mathbb{E}[(Y - y)^+]$ by Jensen; $c(y) \leq \mathbb{E}[Y \mathbf{1}\{Y > y\}] \rightarrow 0$ by dominated convergence.

(iii) A butterfly’s price is a second difference of a convex function, positive by (ii); strictness because convexity is strict. Parity: $c(k) - P(k) = \mathbb{E}[(Y - e^k)^+ - (e^k - Y)^+] = \mathbb{E}[Y - e^k] = 1 - e^k$.

(iv) In log-strike, $\partial^2 c / \partial k^2 = y^2 \partial^2 c / \partial y^2 + y \partial c / \partial y = e^k (e^k f_Y(e^k) - \mathbb{P}(Y > e^k))$. As $k \rightarrow -\infty$, $e^k f_Y(e^k) = f_X(k) \rightarrow 0$ while $\mathbb{P}(Y > e^k) \rightarrow 1$, so the bracket is eventually negative: c is concave in k deep in the money. The computation used only the existence of a density of X vanishing at $-\infty$, so this holds for essentially every arbitrage-free slice in the sense of definition 2.1, not specially for LQD. $\square$

Point (iv) is recorded because it once cost a real audit an afternoon: a convexity check written in log-strike will flag every correct slice as arbitrageable in the deep left wing. Validity checks belong in the strike variable y , where definition 2.1 lives. Figure 5 illustrates (ii) and

(iii) on a fitted real-market slice: the call curve sits below its secant chords everywhere, and finite-width butterflies priced off the fitted slice recover the model density as their width shrinks — (4) made visible.

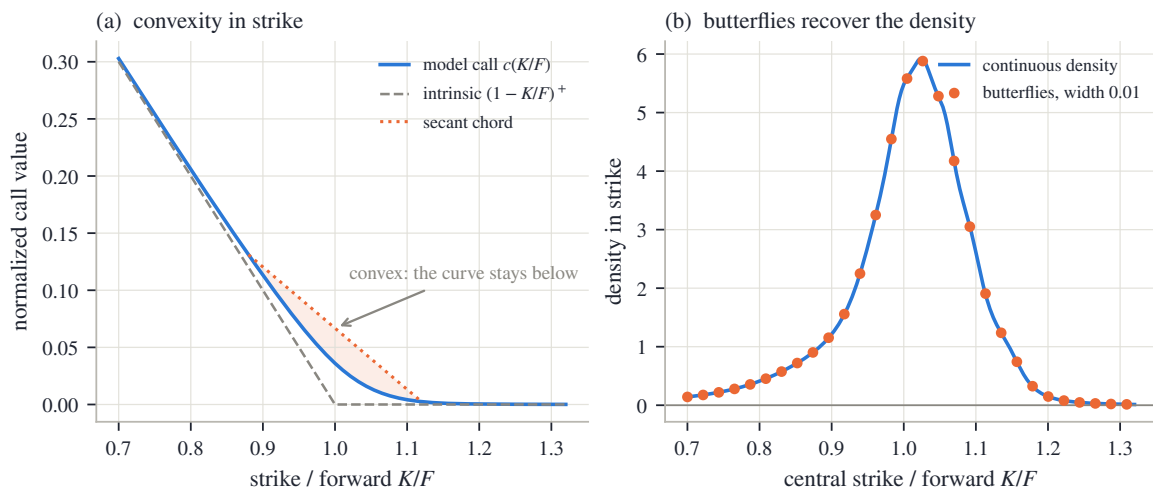


Figure 5: Validity as geometry, on a fitted real-market slice (SPY, December 2026, from the frozen snapshot). Panel A: the normalized call curve against strike $y = e^k$ with a secant chord drawn — the curve is convex, every chord sits above it, so every butterfly has non-negative price. Panel B: discrete butterflies of fixed half-width, priced off the fitted slice and divided by the squared half-width, against the model’s own density (line): the Breeden–Litzenberger identity (4) realized by actual portfolios, to a worst relative gap of 1.49%. Butterfly-freedom is not something this fit achieved; it is something the representation cannot lose.

5.2 The wall is a fact, not a preference

Proposition 5.2 (Finite-forward wall). *Within the family, the normalizing integral converges — and hence the slice exists — if and only if $\lambda_+ < 1$ (proposition 4.2). Moreover the wall is model-free: any law of X whose right tail satisfies $\mathbb{P}(X > x) \geq \kappa e^{-x}$ for some $\kappa > 0$ and all large x has $\mathbb{E}[e^X] = \infty$. No parametrization can price a forward from such a law.*

Proof. The first claim is proposition 4.2. For the second, integrate by parts, or directly: for any level x_0 , $\mathbb{E}[e^X] \geq \mathbb{E}[e^X \mathbf{1}\{X > x_0\}] = \int_0^\infty \mathbb{P}(e^X \mathbf{1}\{X > x_0\} > t) dt \geq \int_{e^{x_0}}^\infty \mathbb{P}(X > \log t) dt \geq \kappa \int_{e^{x_0}}^\infty t^{-1} dt = \infty$. $\square$

In words: λ_+ is the exponential scale of the right return tail (section 6), and a right tail at scale ≥ 1 in X is a return distribution whose mean does not exist. The wall $\lambda_+ < 1$ is the price of a finite forward, charged by probability theory, not by the model. There is no corresponding left-hand wall: λ_- may be arbitrarily large, because $e^X \leq 1$ on the left half of the law’s support contributes at most 1 to the mean.

5.3 What the coordinates cover, and what they do not

The title of this section would be overclaimed by “coordinates for the space of smiles”. The honest statement is a density result *within a class*, with the exclusions named.

Proposition 5.3 (Universality, with exclusions). *Let X^* be a mean-one law whose quantile function Q^* is differentiable with*

$$g^*(u) = \log Q^{*'}(u) + \log u + \log(1 - u)$$

extending continuously to $[0, 1]$ — equivalently, a law with a positive continuous density and exponential tails, of scales $\lambda_-^* = e^{g^*(0)} > 0$ and $\lambda_+^* = e^{g^*(1)} < 1$. Then:

- (i) (approximation) there are admissible LQD slices of increasing order N whose $g_N \rightarrow g^*$ uniformly on $[0, 1]$; their quantiles converge to Q^* uniformly on compact subsets of $(0, 1)$, their normalized call curves converge uniformly in k , and the laws of the gross returns converge in Wasserstein- p distance for every $p \in [1, 1/\lambda_+^*)$ — for laws on $\mathbb{R}$ this metric has the concrete form $W_p(\mu, \nu)^p = \int_0^1 |Q_\mu(u) - Q_\nu(u)|^p du$, the p -th mean cost of carrying one law onto the other rank by rank, which is why it is the natural metric for a quantile-side model;
- (ii) (exclusions) a law with an atom, an interval of zero density inside its support, bounded support, point mass at zero (default), or a Gaussian or faster-decaying tail is not in the family at any finite order. Moreover such laws are unreachable in the limit, in the following precise sense. Let admissible slices have log-speeds g_N converging uniformly on $[0, 1]$; the limit g_∞ is then continuous with $g_\infty(1) \leq 0$, and two branches exhaust the possibilities. If $g_\infty(1) < 0$, the laws converge weakly and the limit belongs to the class of (i). If $g_\infty(1) = 0$, weak convergence can fail outright — the martingale shifts can diverge and the mass escape to $-\infty$ — and if the laws do converge weakly, the limit is a boundary law at the wall: it still has a strictly positive continuous density on all of $\mathbb{R}$ (no atom, no gap, no bounded support, no Gaussian-or-faster tail); what is lost is only the tail scale — its moment strip collapses to $r \leq 1$, and by Fatou it need not even retain unit mean. In neither branch is an excluded law reached.

The proof is in appendix B; the mechanism deserves three sentences here. Approximation works because admissibility is an *open* condition: a Weierstrass polynomial approximation g_N of g^* has $g_N(1) \rightarrow g^*(1) < 0$, so g_N is itself admissible for N large — the family is stable under the approximation that fills it out, which is precisely what a closed condition (an equality constraint, say) would ruin. The exclusions are equally structural: an atom is a jump of the quantile function, a gap is a flat of it, and both are impossible for $Q = x \circ \text{logit}$ with x' a positive continuous exponential of a polynomial, at any order; a Gaussian tail forces $g^*(u) \rightarrow -\infty$ at the endpoint, off the chart entirely.

For desk use the exclusions matter as approximation targets — the double-hump example of section 13 fits a genuinely bimodal density and measures what a finite N delivers — and as honest limits: a pinned settlement print (an atom) or a hard default mass will be *smoothed*, not represented, and the fitter should say so rather than pretend otherwise. Section 15 returns to this contract.

The scope sentence of the paper, then, earned rather than asserted: the LQD construction provides *unconstrained coordinates for a flexible exponential-tail class of mean-one laws* — all of that class, nothing outside it, and the one inequality that delimits it is removed by the logistic chart of section 12.

6 Tails and moments

Extrapolation is where smile models die, so the tails deserve their own section — both for what the model guarantees (exponential tails, Lee-consistent wings, in closed form) and for what it does not (the fitted tail parameters are priors, weakly identified by any realistic strip; the section ends on that).

6.1 Endpoint speeds are tail scales

Proposition 6.1 (Exponential tails). *For every $\theta \in \Theta_N$ there are constants $K_{\pm} > 0$ with*

$$\mathbb{P}(X > x) \sim K_+ e^{-x/\lambda_+} \quad (x \rightarrow +\infty), \quad \mathbb{P}(X < x) \sim K_- e^{x/\lambda_-} \quad (x \rightarrow -\infty), \quad (22)$$

where $\sim$ denotes asymptotic equivalence (ratio $\rightarrow 1$). In return terms: $\mathbb{P}(Y > y) \sim K_+ y^{-1/\lambda_+}$ and $\mathbb{P}(Y < y) \sim K_- y^{1/\lambda_-}$ — exact power tails, hence regularly varying with trivial slowly varying part.

Proof. Take the right tail; the left is a mirror image. By (18), $x(z) = (m+b_+) + \lambda_+ z + \varepsilon(z)$ with $\varepsilon(z) = O(e^{-z})$ — note the *shifted* constant $m+b_+$, since $x = m + \bar{x}$. Now $\mathbb{P}(X > x) = 1 - \Lambda(z(x))$ where $z(x)$ is the inverse transport. Invert the asymptote explicitly: evaluating the transport identity at $z = z(x)$,

$$x = (m+b_+) + \lambda_+ z(x) + \varepsilon(z(x)) \quad \implies \quad z(x) = \frac{x - (m+b_+)}{\lambda_+} - \frac{\varepsilon(z(x))}{\lambda_+} = \frac{x - (m+b_+)}{\lambda_+} + O(e^{-z(x)}),$$

the error keeping its exponential form because it is the same ε , merely moved to the other side of the equation and divided by the constant λ_+ . With $1 - \Lambda(z) = e^{-z}/(1 + e^{-z}) \sim e^{-z}$,

$$\mathbb{P}(X > x) \sim e^{-z(x)} = e^{-(x-m-b_+)/\lambda_+} e^{O(e^{-z(x)})} \sim e^{(m+b_+)/\lambda_+} e^{-x/\lambda_+},$$

which is (22) with $K_+ = e^{(m+b_+)/\lambda_+}$. The return statements substitute $x = \log y$. $\square$

The endpoint speeds are therefore genuine tail *scales*: λ_+ is the number of log-return units the right tail needs to thin by a factor e . This is the sense in which the two leading coordinates of the model are readable. The tails are heavier than log-normal — exponential in X , power-law in Y — which is the empirically comfortable side to err on for equity underliers, and the reason the smiles of fig. 1 turn convexly upward in the wings.

6.2 The moment strip

Moments follow from one change of variables, done once, slowly, because the entire admissibility story drops out of it. Any expectation is a rank integral, and in log-odds the rank measure carries the tent-shaped weight $\rho(z) \asymp e^{-|z|}$:

$$\mathbb{E}[h(X)] = \int_0^1 h(Q(u)) \, du = \int_{-\infty}^{\infty} h(x(z)) \rho(z) \, dz. \quad (23)$$

Proposition 6.2 (Moment strip). *For every $\theta \in \Theta_N$,*

$$\mathbb{E}[e^{rX}] < \infty \quad \iff \quad -\frac{1}{\lambda_-} < r < \frac{1}{\lambda_+}. \quad (24)$$

Equivalently, in return moments: the last finite positive moment order of Y beyond the first is $r_+^ = \sup\{r \geq 0 : \mathbb{E}[Y^{1+r}] < \infty\} = 1/\lambda_+ - 1$, and the last finite inverse moment order is $r_-^* = \sup\{r \geq 0 : \mathbb{E}[Y^{-r}] < \infty\} = 1/\lambda_-$.*

Proof. Apply (23) with $h(x) = e^{rx}$. On the right, $e^{rx(z)}$ grows like $e^{r\lambda_+ z}$ (by the asymptotics in the proof of proposition 6.1) against the weight $\rho(z) \sim e^{-z}$: the integrand behaves like $e^{(r\lambda_+-1)z}$, integrable at $+\infty$ iff $r\lambda_+ < 1$; at equality the integrand tends to a positive constant times 1 and diverges. On the left, $e^{rx(z)} \asymp e^{r\lambda_- z}$ against $\rho(z) \sim e^z$: integrable at $-\infty$ iff $r\lambda_- > -1$. Both endpoint failures are exact ties between the exponential growth of the integrand and the exponential decay of the rank weight. The return statements are the substitutions $\mathbb{E}[Y^{1+r}] = \mathbb{E}[e^{(1+r)X}]$ (finite iff $1+r < 1/\lambda_+$) and $\mathbb{E}[Y^{-r}] = \mathbb{E}[e^{-rX}]$ (finite iff $r < 1/\lambda_-$). $\square$

One point of the strip is load-bearing: the martingale integral is the case $r = 1$, and $1 < 1/\lambda_+$ is exactly the wall $\lambda_+ < 1$ of proposition 5.2 — the same fact seen for the third time, now as “the forward is the first moment, and the first moment must be finite”. No left-hand condition arises because $1 > -1/\lambda_-$ always.

6.3 Lee’s formula, and the closed speed-to-slope maps

Lee’s moment formula [16] is the model-free bridge from moments to wing geometry. Writing $\beta_R = \limsup_{k \rightarrow +\infty} w(k)/k$ and $\beta_L = \limsup_{k \rightarrow -\infty} w(k)/|k|$ for the asymptotic slopes of the total-variance smile, Lee proved

$$\beta_R = \Psi(r_+^*), \quad \beta_L = \Psi(r_-^*), \quad \Psi(r) = 2 - 4(\sqrt{r^2 + r} - r) \in [0, 2], \quad (25)$$

with $r_\pm^*$ the critical moment orders of proposition 6.2 (and $\Psi(\infty) = 0$, $\Psi(0) = 2$: the fewer finite moments, the steeper the admissible wing, capped at slope 2). For LQD the critical orders are explicit, so the composition collapses to closed form.

Proposition 6.3 (Speed-to-slope maps). *For an LQD slice with endpoint speeds $\lambda_\pm$,*

$$\beta_R = \Psi\left(\frac{1}{\lambda_+} - 1\right) = \frac{2\lambda_+}{(1 + \sqrt{1 - \lambda_+})^2}, \quad \beta_L = \Psi\left(\frac{1}{\lambda_-}\right) = \frac{2\lambda_-}{(1 + \sqrt{1 + \lambda_-})^2}. \quad (26)$$

Both maps are increasing; $\beta_R \rightarrow 2$ as $\lambda_+ \rightarrow 1$ (the Lee ceiling is reached exactly at the wall) and $\beta_\pm \approx \lambda_\pm/2$ for small speeds. Moreover, the limsups in (25) are genuine limits for LQD slices.

Proof. Right wing: put $r = 1/\lambda_+ - 1$, so $r^2 + r = r(r + 1) = (1 - \lambda_+)/\lambda_+^2$ and $\sqrt{r^2 + r} = \sqrt{1 - \lambda_+}/\lambda_+$. With $s = \sqrt{1 - \lambda_+}$,

$$\Psi(r) = 2 - \frac{4}{\lambda_+}(s - s^2) = 2 - \frac{4s(1 - s)}{1 - s^2} = 2 - \frac{4s}{1 + s} = \frac{2(1 - s)}{1 + s} = \frac{2(1 - s^2)}{(1 + s)^2} = \frac{2\lambda_+}{(1 + s)^2}.$$

Left wing: put $r = 1/\lambda_-$, so $\sqrt{r^2 + r} = \sqrt{1 + \lambda_-}/\lambda_-$, and with $t = \sqrt{1 + \lambda_-}$,

$$\Psi(r) = 2 - \frac{4}{\lambda_-}(t - 1) = 2 - \frac{4(t - 1)}{t^2 - 1} = 2 - \frac{4}{1 + t} = \frac{2(t - 1)}{1 + t} = \frac{2(t^2 - 1)}{(1 + t)^2} = \frac{2\lambda_-}{(1 + t)^2}.$$

Monotonicity and the endpoint values are read off the final expressions. The small-speed expansions follow from $s, t \rightarrow 1$. For the limit claim: Lee’s theorem gives only limsups in general, but Benaïm and Friz [1] showed that when the return tail is regularly varying — as proposition 6.1 establishes here, with exact power tails — the wing slope converges as an ordinary limit. $\square$

The endpoint coefficients therefore do double duty: they regularize the extrapolation *and* identify the entire moment budget and both asymptotic wing slopes, in closed form, from two fitted numbers. Figure 6 draws the three-link chain — endpoint speed to last finite moment to Lee slope — with the fitted values of two real underliers marked on each link: for the SPY December 2026 slice of section 13, $\lambda_- = 0.432$, $\lambda_+ = 0.066$, $\beta_L = 0.179$, $\beta_R = 0.034$; for the long NVDA node (December 2027, the 1.37-year slice of fig. 12), $\lambda_- = 1.421$, $\lambda_+ = 0.225$, $\beta_L = 0.435$, $\beta_R = 0.128$. Single-name tails are visibly heavier than index tails, as they should be — and the NVDA left scale is a live illustration of the wall’s asymmetry: $\lambda_- > 1$ is perfectly admissible (only the *right* scale is walled, proposition 5.2); its price is a last finite inverse-moment order $1/\lambda_- = 0.70$ below one, i.e. even $\mathbb{E}[1/Y]$ diverges, while the forward stays finite.

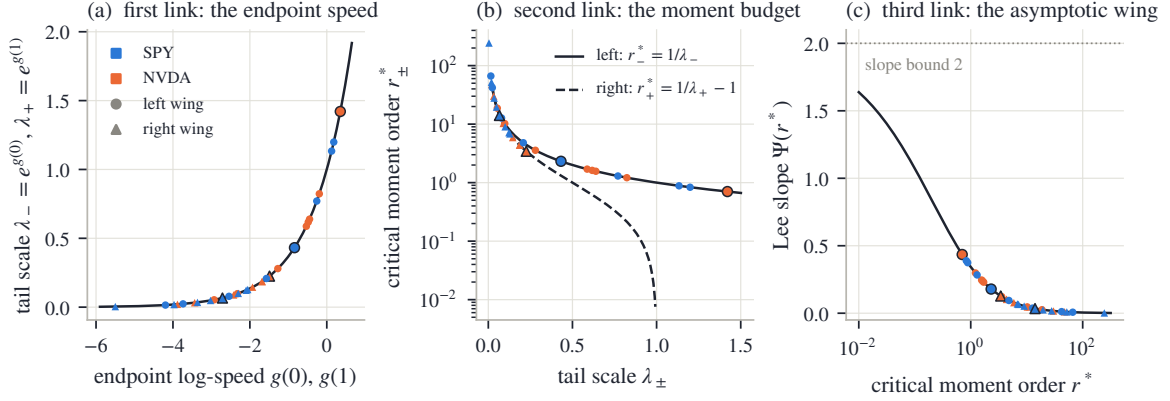


Figure 6: The tail chain in three panels. Panel A: endpoint speed λ to last finite moment order, $1/\lambda_-$ on the left tail and $1/\lambda_+ - 1$ on the right (24); the finite-forward wall appears as the vertical asymptote of the right-tail map at $\lambda_+ = 1$, where the right tail’s moment budget is exhausted — the wall belongs to the right scale only, and the marked NVDA left scale $\lambda_- > 1$ is legal. Panel B: the Lee map Ψ (25). Panel C: the composed speed-to-slope maps (26), with the model-free ceiling $\beta = 2$ dotted. Markers on all three panels: the fitted SPY December 2026 and NVDA December 2027 values from the frozen snapshot — equity fits use a small fraction of the admissible range, sitting far below the ceiling.

6.4 The honesty clause: the limit is not the wing

Two cautions convert this section from advertisement to information.

First, the asymptotic slope is not a quote at any tradable strike — and at tradable strikes it can err in *either direction*. Near the money the effective variance slope $w(k)/|k|$ is dominated by $w_0/|k|$ and diverges; far out it tends to β ; but in between its path is governed by the wing’s intercept — writing the wing as $w(k) \approx \beta|k| + \text{const}$, the effective slope approaches the limit from above when the constant is positive and from *below* when it is negative, and nothing pins the sign. Both cases occur on one real slice. Figure 7 plots the fitted SPY December node: on the call wing the effective slope sits above its limit, ratio 1.69 at the 10Δ strike, still 1.05 at 1Δ — an overshoot; on the put wing it *undershoots*, ratio 0.70 at 10Δ and 0.70 at 1Δ , both below one. The curves in the figure are truncated where the option price falls below 10^{-13} of forward — beyond that, implied variance is Black-inversion noise — so the asymptote has not been approached at any strike a computer, let alone a market, can quote. The conclusion is therefore stronger than a one-sided warning: a desk consuming $\beta_{\pm}$ should read them as extrapolation boundary conditions and never price a wing off them directly, because at ten-delta the limit overstates one wing of this very slice and understates the other.

Second — and this is the information clause of the paper’s validity/information/convention triad — the fitted $\lambda_{\pm}$ are *priors dressed as parameters*. No finite quoted strip determines an asymptotic decay rate; what a fit reports as λ_+ is what the quotes, the basis order, the ridge, the vega weighting, and the endpoint coupling *jointly* selected. The model’s contribution is to make that selection explicit, low-dimensional, and overridable (a desk can pin $\lambda_{\pm}$ through the endpoint chart of section 12), not to conjure tail information the market did not supply. Section 9 returns to identifiability with the fitting machinery on the table.

7 The ledger

This section introduces the paper’s central computational object and derives, from it alone, every pricing formula the rest of the paper uses.

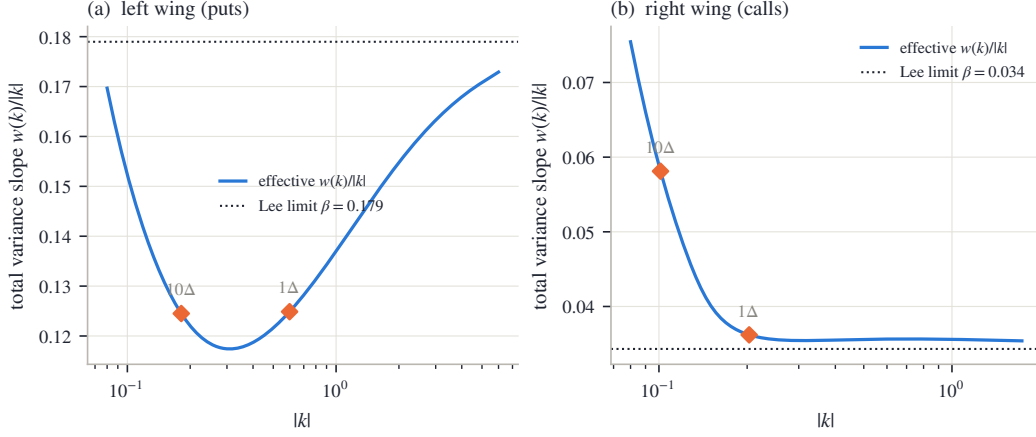


Figure 7: The limit is not the wing — in either direction. Effective total-variance slope $w(k)/|k|$ against $|k|$ (log scale) for the fitted SPY December 2026 slice, both wings; the Lee limits $\beta_{\pm}$ of (26) dotted; diamonds mark the 10Δ and 1Δ strikes. The call wing (right panel) sits *above* its limit — ratios 1.69 at 10Δ , 1.05 at 1Δ ; the put wing (left panel) dips *below* its limit and approaches it from underneath — ratios 0.70 and 0.70. Curves are truncated where the option price drops below 10^{-13} of forward, beyond which implied variance is inversion noise. The asymptotic slope is a boundary condition on extrapolation, not a quote at a tradable strike.

7.1 Definition and first properties

Definition 7.1 (Upper-share ledger). For an admissible slice, define

$$G(z) = \int_z^\infty e^{x(t)} \rho(t) dt = \mathbb{E}[Y \mathbf{1}\{Z > z\}]. \quad (27)$$

$G(z)$ is the share of the (unit) mean carried by outcomes above the z -th log-odds level — hence the name. Three properties are immediate and get used constantly:

$$G(-\infty) = \mathbb{E}[Y] = 1, \quad G(+\infty) = 0, \quad G'(z) = -e^{x(z)} \rho(z) < 0. \quad (28)$$

The ledger is a smooth, strictly decreasing curve from 1 to 0, whose derivative is explicitly known at every node of whatever grid carries it. That last fact looks small and is not: it is what will make interpolation fourth-order accurate without any spline solve (section 12), and what makes the cancellation theorem of section 10 one line.

For a strike k , define the *strike root* z_k and its rank u_k :

$$x(z_k) = k, \quad u_k = \Lambda(z_k) = F_X(k), \quad (29)$$

unique because x is a strictly increasing bijection of $\mathbb{R}$.

7.2 Pricing identities

Proposition 7.2 (Ledger pricing). For every admissible slice and every k :

$$c(k) = G(z_k) - e^k(1 - u_k), \quad (30)$$

$$P(k) = e^k u_k - (1 - G(z_k)), \quad (31)$$

$$c'(k) = -e^k(1 - u_k), \quad (32)$$

$$c''(k) = -e^k(1 - u_k) + e^k \frac{\rho(z_k)}{x'(z_k)}, \quad (33)$$

$$f_X(k) = e^{-k}(c''(k) - c'(k)) = \frac{\rho(z_k)}{x'(z_k)}. \quad (34)$$

Proof. (30): split the payoff on the event $\{Y > e^k\} = \{Z > z_k\}$ (the events coincide because x is increasing),

$$c(k) = \mathbb{E}[(Y - e^k)\mathbf{1}\{Z > z_k\}] = \mathbb{E}[Y\mathbf{1}\{Z > z_k\}] - e^k \mathbb{P}(Z > z_k) = G(z_k) - e^k(1 - u_k) :$$

the asset delivered on exercise, minus the strike bill paid on the same event — a call price is the difference of two ledgers, both monotone, both already on the grid. (31) is the same split on the complementary event, and subtracting gives parity $c - P = G(z_k) + (1 - G(z_k)) - e^k = 1 - e^k$, as it must.

(32): differentiate (30) in k . The chain rule through the moving root contributes $[G'(z_k) + e^k \rho(z_k)] \frac{dz_k}{dk}$; but $G'(z_k) = -e^{x(z_k)} \rho(z_k) = -e^k \rho(z_k)$ by (28) and (29), so the bracket vanishes identically, and only the explicit $-e^k(1 - u_k)$ term survives. (This vanishing bracket is the envelope mechanism that theorem 10.1 elevates to a theorem: the root sits exactly where the integrand of the combined asset-and-cash ledger is zero, so moving it is free to first order.)

(33): differentiate (32), now keeping the root term, with $dz_k/dk = 1/x'(z_k)$ and $\Lambda' = \rho$:

$$c''(k) = -e^k(1 - u_k) + e^k \rho(z_k) \frac{dz_k}{dk} = -e^k(1 - u_k) + e^k \frac{\rho(z_k)}{x'(z_k)}.$$

(34): subtract (32) from (33) and multiply by e^{-k} ; the result $\rho(z_k)/x'(z_k)$ is $f_X(k)$ by (14). This is Breeden–Litzenberger (4) transported to log-strike: $e^{-k}(c'' - c') = f_X(k)$, equivalently $\partial^2 c / \partial y^2 = f_Y(y)$ at $y = e^k$. $\square$

Equation (34) deserves its plain-language reading: on the grid, the risk-neutral density is *one division* of two arrays the build already holds, ρ and $x' = e^{g \circ \Lambda}$. No differentiation of prices, no finite differences, no smoothing: the density was never anything but these two arrays, and the call curve was derived *from* it, not the other way around. This is the practical content of choosing coordinates on laws rather than on curves.

Two further remarks. First, (32) identifies the *digital*: $-e^{-k}c'(k) = 1 - u_k = \mathbb{P}(Y > e^k)$, so the model's exercise probabilities are read off the same root. Second, all five identities hold for every admissible θ — they are structure, not calibration; the optimizer of section 9 prices every trial point through (30) and nothing else.

A first ticket, on the solvable slice. The identities deserve one concrete run before they carry a real market. Take the constant-speed slice of section 4.3, with the scale root-solved so that the ATM volatility is exactly 20% at six months: $s = 0.081250$, shift $m = -0.010883$ in closed form. This is the slice drawn in fig. 1, and its two marked strike roots are now priceable. At the money, the forward's rank is 53.34% — *above* one half: the shift $m < 0$ pushes the median return below zero so that e^X can average to one, which is Jensen's inequality read off a percentile. For the $k = 0.10$ call: the strike's rank is $u_k = 79.65\%$, the ledger holds $G(z_k) = 0.247221$ of the forward above that rank, the strike bill on the same event is $e^k(1 - u_k) = 0.224876$, and (30) gives

$$c(0.10) = 0.247221 - 0.224876 = 0.022345 \quad \implies \quad \sigma_{\text{imp}} = 20.60\%,$$

about 0.6 volatility points above the 20% ATM level: the first taste of a smile, produced by a symmetric law whose exponential tails out-fatten the log-normal's. Section 13 repeats this exercise on a real listed option.

7.3 The variance swap as the build's first moment

The log contract gives the standard model-free variance proxy [5, 7, 17]: for a continuous forward path monitored continuously, the fair strike of a variance swap over $[0, T]$ is

$$w_{\text{vs}} = -2 \mathbb{E}[\log Y] = -2 \mathbb{E}[X] = -2 \int_{-\infty}^{\infty} x(z) \rho(z) dz, \quad (35)$$

one more integral over the arrays of the same build, with integrand decaying like $|z|e^{-|z|}$. Where the classical route replicates the log contract from a strip of option prices across strikes [7], here the distribution is available directly, so the strip replication is bypassed — (35) is exact for the fitted law, and its parameter derivative rides the Jacobian pass of section 10 at no extra build.

The honesty clause attaches immediately: (35) prices the *log contract*. Its identification with a realized-variance swap assumes continuous monitoring of a continuous path; jumps and discrete sampling take the usual corrections, which are outside the scope of a single-expiry static model. The paper therefore reports $\sigma_{\text{vs}} = \sqrt{w_{\text{vs}}/\tau}$ as “the var-swap strike of the fitted law” and claims nothing about realized variance.

7.4 Five uses of one array

It is worth pausing to collect the thread, because the pattern is the paper’s argument in miniature. One quadrature build produces the arrays $x(z_j)$, $x'(z_j)$, $G(z_j)$, $G'(z_j)$ on a fixed grid. From these: pricing is (30) — two lookups and a root; the density is (34) — one division; the variance swap is (35) — one dot product; the calibration Jacobian is the nodal parameter-sensitivity of G evaluated at the same roots — theorem 10.1; and calendar order across expiries is the pointwise comparison of two such ledgers — theorem 11.3. Five computations, one array. A risk system built on this model holds one object per slice and interrogates it; nothing is ever re-derived from prices, because nothing was ever derived from prices in the first place.

8 The at-the-money microscope

A desk reads a smile through three numbers: the at-the-money volatility, its slope in log-moneyness (the skew), and its curvature. For most parametrizations these are obtained by fitting, plotting, and finite differencing. For the LQD slice they are *exact closed-form functionals* of three numbers read off the law at a single point, and the skew formula carries an interpretation worth the whole section.

Throughout, primes on c , w , σ denote k -derivatives evaluated at $k = 0$; the three law-side inputs are

$$c_0 = c(0) = G(z_*) - (1 - u_*), \quad u_* = \Lambda(z_*), \quad f_* = f_X(0) = \frac{\rho(z_*)}{x'(z_*)}, \quad (36)$$

where $z_* = z_0$ is the root $x(z_*) = 0$: the ATM call, the *rank of the forward*, and the ATM return density. By (32)–(33), $c'(0) = -(1 - u_*)$ and $c''(0) = f_* - (1 - u_*)$.

Proposition 8.1 (Exact ATM handles). *Let w_0 solve $2\Phi(\sqrt{w_0}/2) - 1 = c_0$, i.e. $w_0 = 4[\Phi^{-1}(\frac{1+c_0}{2})]^2$, and write $d = \sqrt{w_0}/2$ and*

$$\Delta = u_* - \Phi(d) \quad (37)$$

for the digital mismatch. Then the total-variance smile $w(k)$ defined by $B(k, w(k)) = c(k)$ satisfies, at $k = 0$,

$$w(0) = w_0, \quad w'(0) = \frac{2\sqrt{w_0}}{\varphi(d)} \Delta, \quad (38)$$

$$w''(0) = \frac{2\sqrt{w_0}}{\varphi(d)} f_* - 2 + \left(\frac{1}{2w_0} + \frac{1}{8}\right) w'(0)^2, \quad (39)$$

and in volatility units, with $\sigma(k) = \sqrt{w(k)}/\tau$,

$$\sigma(0) = \sqrt{\frac{w_0}{\tau}}, \quad \sigma'(0) = \frac{\Delta}{\varphi(d)\sqrt{\tau}}, \quad \sigma''(0) = \frac{1}{\sqrt{\tau}} \left[\frac{f_*}{\varphi(d)} - \frac{1}{\sqrt{w_0}} + \frac{\sqrt{w_0}}{4} \left(\frac{\Delta}{\varphi(d)} \right)^2 \right]. \quad (40)$$

The derivation is implicit differentiation of the identity $B(k, w(k)) = c(k)$ through the Black formula, using the five ATM Black partials; it is mechanical and two pages long, so it lives in appendix B. What belongs here is the one-line heart of it. Differentiating once, $\partial_k B + \partial_w B w'(0) = c'(0)$, with $\partial_k B = -(1 - \Phi(d))$ and $\partial_w B = \varphi(d)/(2\sqrt{w_0})$ at the money; solving,

$$w'(0) = \frac{c'(0) - \partial_k B}{\partial_w B} = \frac{-(1 - u_*) + (1 - \Phi(d))}{\varphi(d)/(2\sqrt{w_0})} = \frac{2\sqrt{w_0}}{\varphi(d)}(u_* - \Phi(d)). \quad (41)$$

The skew is a digital mismatch. Equation (41) says something that deserves plain language, and the reading must be established with signs in hand rather than by rhetoric. $-e^{-k}c'(k)$ is the price of the digital call — the probability of finishing above the strike — so $1 - u_*$ is the *model's* at-the-money digital, while $1 - \Phi(d)$ is the digital of the flat-smile Black model matched to the same ATM price. Their difference is exactly the negative of (37):

$$\Delta = u_* - \Phi(d) = (1 - \Phi(d)) - (1 - u_*) = (\text{Black digital}) - (\text{model digital}),$$

and by (38) the total-variance skew is the positive factor $2\sqrt{w_0}/\varphi(d)$ times Δ . Three sign checks pin the orientation. *The null case:* if the law is exactly the matched Black lognormal, $X \sim \mathcal{N}(-w_0/2, w_0)$, then $u_* = \mathbb{P}(X \leq 0) = \Phi(\sqrt{w_0}/2) = \Phi(d)$ identically, so $\Delta = 0$ and the smile is flat at the money — the formula contains its own benchmark. *The skewed case:* a fitted equity-index slice tilts down to the right, $\sigma'(0) < 0$, so by (40) $\Delta < 0$: the model's ATM digital is *rich* relative to flat-Black, i.e. the forward sits at a *lower* model rank than the matched flat-Black percentile, $u_* < \Phi(d)$. The distributional mechanism is the crash tail: the law buys its fat left tail by parking the compensating probability just above the forward, so more outcomes finish above the forward — and fewer below — than the matched lognormal would allow. (Note what is *not* claimed: u_* may sit on either side of one half — $\Phi(d) > \frac{1}{2}$ always, and $\Delta < 0$ only places u_* below $\Phi(d)$; the fitted SPY node of section 13 has u_* below one half as well, while the symmetric slice of section 7 has u_* above one half with Δ near zero.) *The general rule:* $\text{sign } \sigma'(0) = \text{sign } \Delta$ — an upward tilt is exactly a cheap model digital, a downward tilt exactly a rich one. The skew is not a shape parameter of a curve; it is a statement about where the forward sits in the distribution. The curvature (39) adds exactly one more distributional fact, the ATM density f_* . Level, digital, density: three numbers of the law, three handles of the smile, no asymptotic expansion and no finite difference anywhere.

Two practical notes close the section. First, exactness matters operationally: the handles feed risk systems and cross-expiry logic, and a handle computed by finite-differencing a plotted curve inherits the plotting grid; (38)–(40) inherit only the root solve, which is polished to machine precision (section 12). Second, the identities run in both directions: given target handles $(\sigma(0), \sigma'(0), \sigma''(0))$ one can solve the three equations for (c_0, u_*, f_*) and use them as calibration targets — the reference implementation exposes this as a local chart around a fitted slice whose first three coordinates *are* the handles, built from the Jacobian of section 10 (appendix A); the paper does not develop that machinery further.

9 Information: calibrating one expiry

Eight sections built a space in which every point is an arbitrage-free slice. Choosing the right point is now a plain nonlinear least-squares problem — which was the design goal — and the interesting questions are informational: what do the quotes determine, what does the objective choose, and how are the two kept from masquerading as one another.

9.1 The objective

Given quotes σ_i at log-moneyness k_i for one expiry, with total variances $w_i = \sigma_i^2 \tau$, the calibrator solves

$$\min_{\theta} \sum_i \left(\frac{c(k_i; \theta) - B(k_i, w_i)}{\partial_{\sigma} B(k_i, w_i) + \eta} \right)^2 + \alpha \sum_{n \geq 4} n^{2\nu} a_n^2, \quad (42)$$

an unconstrained problem over the chart coordinates of section 12, solved by a trust-region least-squares method [3, 22]. Three design decisions are baked into (42), each with a reason.

Prices, not volatilities, in the residual. The model's trial point is always a genuine law, so its prices always exist; inverting them to implied volatility inside the loop would add a root-solve per quote per iterate and would fail exactly on the wild trial points a trust-region method is entitled to probe. Fitting in price space keeps every iterate well-defined.

Vega normalization. Dividing the price error by the Black vega $\partial_{\sigma} B = \varphi(d_+) \sqrt{\tau}$ at the quote restores, to first order, a volatility error — so the objective's units are the units the desk thinks in, and residuals of different strikes are commensurable. The floor η (a small constant, table 4) matters in the far wings, where vega vanishes: below the floor, the residual degrades gracefully from a vol error to a scaled price error. This is a deliberate trade — a vol metric at zero vega amplifies noise without bound — but it means short-dated wing quotes are fitted under a price-error regime, and a reader of such fits should know it.

An explicit ridge, sparing the workhorses. The term $\alpha \sum_{n \geq 4} n^{2\nu} a_n^2$ (defaults in table 4; $\nu = 1$) suppresses high-order oscillation where quotes are sparse. The first two body modes a_2, a_3 — smile convexity and asymmetry, which liquid quotes always pin — are deliberately unpenalized. The ridge is a *convention*, disclosed as such: it is the tie-breaker wherever the likelihood is flat, and section 6 already identified the place the tie-breaking shows up (the tail parameters).

One more residual row rides along in every fit: a smooth barrier $\log(1 + e^{50(\lambda_+ - 0.90)})$, negligible below $\lambda_+ \approx 0.9$ and rising steeply toward the wall. In the production chart the wall is unreachable (section 12), so the row is not a feasibility device; it is an economic prior against fits that quote a nearly infinite forward-moment budget, and it is listed here because it is part of the objective, not of the geometry. Its units are frankly conventional: the row is dimensionless with unit weight, and its constants are sized so that it is invisible below the center and reaches order one — comparable to a multi-vol-point data residual — within a few hundredths of λ_+ beyond it; center and steepness are listed with the other disclosed conventions in table 4. A related smoothness note belongs here: the hinge residuals of (43) have a kink at the band edge, but the solver minimizes the sum of *squares*, and $x \mapsto ((x)^+)^2$ is continuously differentiable, so the objective the trust region sees is C^1 at the edge — the stronger $\cos^2$ smoothing applied to the calendar rows of section 11 serves their strike-dependent *weights*, which would otherwise imprint kinks of their own.

9.2 Fit targets: mid, band, haircut

A quote is not a number; it is a bid and an ask. The calibrator supports three targets.

Mid. The residual of (42) against the mid volatility. Right when spreads are tight and symmetric.

Band. Replace the residual by a hinge on the band edges plus a weak mid anchor: writing $[p_i^{\text{lo}}, p_i^{\text{hi}}]$ for the quote's price band (the vol band mapped through Black once, before the solve) and ϕ_i for the vega normalizer of (42), the quote contributes

$$\frac{(c(k_i; \theta) - p_i^{\text{hi}})^+ + (p_i^{\text{lo}} - c(k_i; \theta))^+}{\phi_i} \quad \text{and} \quad \sqrt{0.05} \frac{c(k_i; \theta) - p_i^{\text{mid}}}{\phi_i} : \quad (43)$$

zero cost anywhere inside the band, quadratic cost outside, plus five percent of the mid-fit pull to break ties among interior solutions. Band fitting says honestly that the market quoted an interval.

Haircut (the default for the examples of section 13). Bid-ask bands on listed options are conservative: the touch is often wider than where business is done. The haircut target shrinks each band by a fixed volatility margin h before applying (43):

$$\sigma_i^{\text{lo}} = \min(\sigma_i^{\text{bid}} + h, \sigma_i^{\text{mid}}), \quad \sigma_i^{\text{hi}} = \max(\sigma_i^{\text{mid}}, \sigma_i^{\text{ask}} - h), \quad (44)$$

with h half a volatility point by default. The min/max guards supply the degenerate-band behavior: a quote whose spread is already tighter than $2h$ collapses to $\sigma^{\text{lo}} = \sigma^{\text{hi}} = \sigma^{\text{mid}}$ — a mid fit on that strike — so the haircut can only tighten a band, never invert it. The degeneracy clause is not hypothetical; it is the live state of half the paper’s data. In the frozen snapshot, *none* of the deep-dive SPY node’s bands survives the shrink — the live-band share is 0% — which is exactly the statement that every one of its quoted spreads is tighter than $2h = 100$ vol bp (for color: the median spread is 4 vol bp, an order of magnitude inside the guard). The entire SPY book fits, in effect, to mid, and the SPY figures of section 13 accordingly show raw bid-ask quotes rather than haircut bands. The NVDA book is the other regime — median spreads near $2h$ (100 vol bp on the long node), with 47% and 52% of bands live on the featured nodes — so NVDA is where the band mechanics actually engage. One recipe, two regimes, resolved per quote by (44) with no user decision: that is the intended design.

9.3 What is pinned, and what is chosen

The informational accounting of a fit, stated once, plainly. A liquid strip pins: the body of the distribution between the outermost quotes — equivalently the call curve there — the three ATM handles, the var-swap-style low moments, and the wing behavior *through* the quoted range. The objective chooses: everything beyond the outermost quotes (via the endpoint speeds, i.e. via the basis and the two tail coordinates), the partition of any slack inside bid-ask bands (via the mid anchor), and the smoothness of the body at scales below the quote spacing (via the ridge and the order). The choices are few, explicit, and orthogonal to validity — moving any of them yields a different *admissible* slice. The tail-parameter honesty of section 6 is this accounting applied to $\lambda_{\pm}$; the order-control study of section 13 (fig. 14) is the same accounting applied to N : at matched quote error, the extra order buys density resolution, and the choice of N decides how much of the inter-quote structure the fitter is willing to assert.

9.4 The order guard

Order is a convention with a failure mode, so it is governed. The family at order N has $N + 1$ parameters; a strip of n_q quotes fits at *effective* order

$$N_{\text{eff}} = \max\left(\min(N, \lfloor n_q/2 \rfloor - 1), \min(N, 6)\right), \quad (45)$$

that is: parameters are capped at half the quote count, with a floor at order 6 (or N itself if smaller) so thin strips still fit a usable smile. The principled motivation is *identification*: as the parameter count approaches the quote count, the fit crosses from estimating a curve to interpolating the noise, and the delta-method uncertainty bands on the fitted smile blow up accordingly — the guard keeps the fit on the estimating side. A second, latency-side motivation is real but *book-dependent*, and honesty requires reporting both halves of the evidence. On the snapshot’s own thinnest strip — the 17-quote shortest NVDA node, 2 calendar days from the reference date — deliberately over-parameterizing past the guard produces no distress at all: sweeping the parameter-to-quote ratio through 0.47 / 0.59 / 0.71 costs 14 / 15 /

9 residual evaluations (25 ms, 25 ms, 30 ms): flat. On stressed same-day books, however, the reference implementation has historically observed the same sweep multiply evaluation counts by two orders of magnitude and push a fit from tens of milliseconds to seconds — an error-bar-saturated strip hands the trust region a nearly flat valley to wander, and the failure appears without warning on some books and not others. The guard (45) is therefore cheap insurance whose measured cost on a clean book is nil: it holds the NVDA strip at $N_{\text{eff}} = 7$, and section 13 shows the guarded fit on live data.

9.5 Starting points

Cold starts use the exactly solvable slice of section 4.3: $a_n = 0$, $L = R = \log s$ with $s = \sqrt{3w_0}/\pi$ matched to the ATM-interpolated quote variance — a seed whose bias is known ($\approx 8\%$ low in ATM variance, fig. 3) and gone after one iteration. Surface sweeps warm-start each expiry from its already-fitted shorter neighbor, since adjacent maturities carry nearly the same smile. A warm start is a prior in disguise — around a scheduled event, yesterday’s unimodal seed can shepherd a genuinely bimodal fit into the wrong basin — so the practice is a convention to be disclosed, and multi-start protocols are the audit for it. The audit was run for this paper: 10 randomized starts on each of 2 live nodes (the deep-dive SPY node and the long NVDA node of section 13) found 1 basin per node — worst intra-cluster parameter spread 2.3×10^{-6} , worst spread in fit quality 0.00 vol bp. A measurement on two liquid strips, not a theorem, and stressed or event-laden boards remain exactly where the caution applies.

10 The analytic Jacobian

A least-squares solver spends most of its time on the Jacobian: with $P = N + 1$ parameters, finite-differencing the residuals rebuilds the entire slice $P + 1$ times per iteration. The ledger removes the rebuilds. The key is a cancellation identity, and it is the cleanest piece of mathematics in the implementation.

10.1 The envelope-cancellation theorem

The priced call (30) depends on θ twice: explicitly through the curves x and G , and implicitly through the strike root $z_k(\theta)$ solving $x(z_k; \theta) = k$. A naive gradient must differentiate the root. It need not.

Theorem 10.1 (Envelope cancellation). *For every admissible θ , every strike k , and every coordinate θ_j ,*

$$\frac{\partial c(k; \theta)}{\partial \theta_j} = \frac{\partial G(z; \theta)}{\partial \theta_j} \Big|_{z=z_k(\theta)} : \quad (46)$$

the implicit dependence through the strike root contributes nothing.

Proof. Differentiate $c = G(z_k; \theta) - e^k(1 - \Lambda(z_k))$ in θ_j , writing the root’s motion explicitly:

$$\frac{\partial c}{\partial \theta_j} = \frac{\partial G}{\partial \theta_j} \Big|_{z_k} + \left[G'(z_k) + e^k \rho(z_k) \right] \frac{\partial z_k}{\partial \theta_j}.$$

By (28), $G'(z) = -e^{x(z)}\rho(z)$, and at the root $e^{x(z_k)} = e^k$, so the bracket is $-e^k\rho(z_k) + e^k\rho(z_k) = 0$. $\square$

Remark 10.2. This is an envelope argument. The call is an integral whose lower limit sits exactly where its integrand — the payoff $e^{x(t)} - e^k$ against the rank measure — vanishes: the exercise boundary is, by definition, the point of indifference, so moving it is free to first order. It is the same mechanism by which the early-exercise boundary drops out of American-option sensitivities. The reader may also recognize (32) as the case “ $\theta_j = k$ ” of the same computation.

10.2 One pass for all columns

With the root frozen by theorem 10.1, differentiating the build is mechanical, because g is *affine* in θ : its coordinate derivatives are the fixed basis functions

$$\frac{\partial g}{\partial L} = 1 - u, \quad \frac{\partial g}{\partial R} = u, \quad \frac{\partial g}{\partial a_n} = P_n(1 - 2u),$$

cached once per grid. Write $\bar{x}$ for the unshifted transport and I for the normalizing integral of proposition 4.2. Then, for each coordinate θ_j (all integrals over the fixed grid):

$$\frac{\partial \bar{x}(z)}{\partial \theta_j} = \int_0^z e^{g(\Lambda(t))} \frac{\partial g}{\partial \theta_j}(\Lambda(t)) dt \quad (\text{cumulative pass 1}) \quad (47)$$

$$\frac{\partial m}{\partial \theta_j} = -\frac{1}{I} \int_{-\infty}^{\infty} e^{\bar{x}(z)} \frac{\partial \bar{x}(z)}{\partial \theta_j} \rho(z) dz \quad (\text{one dot product}) \quad (48)$$

$$\frac{\partial G(z)}{\partial \theta_j} = \int_z^{\infty} e^{x(t)} \left(\frac{\partial m}{\partial \theta_j} + \frac{\partial \bar{x}(t)}{\partial \theta_j} \right) \rho(t) dt \quad (\text{cumulative pass 2, from the right}). \quad (49)$$

Every step is a cumulative integral of arrays already in hand, because differentiation commutes with integration and the exponential differentiates into itself times a cached basis column. The integrand of (49), negated, is also the nodal derivative $\partial G'/\partial \theta_j$ — exactly the value/derivative pair the Hermite interpolation of section 12 wants, so the interpolated Jacobian has the same fourth-order accuracy as the interpolated price. The remaining residual rows differentiate in one line each: the ridge is diagonal, the band hinge multiplies (46) by the violation sign, the barrier row is a scalar chain through $\partial \lambda_+/\partial \theta = \lambda_+ \cdot (0, 1, 1, -1, \dots)$, and the var-swap row is

$$\frac{\partial w_{vs}}{\partial \theta_j} = -2 \int_{-\infty}^{\infty} \left(\frac{\partial m}{\partial \theta_j} + \frac{\partial \bar{x}(z)}{\partial \theta_j} \right) \rho(z) dz, \quad (50)$$

riding the same pass — a var-swap-targeted fit costs the same as a plain one.

10.3 Complexity, and the measured speedup

Honesty about what was gained. The analytic pass computes P cumulative sensitivity integrals on one build: cost $O(P n_{\text{grid}})$. The finite difference computes $P+1$ *complete* builds — each with its normalization, exponentials, and root solves — also $O(P n_{\text{grid}})$ in symbols. The analytic route removes the rebuild constant, not the linear-in- P scaling; the honest claim is a constant-factor win, and it is measured, not asserted. On the reference implementation, timed by the protocol of section 12 (median and interquartile range over interleaved analytic/finite-difference solves in a single run, never best-of- n), the analytic Jacobian reaches the same optimum — costs agreeing to 1.7×10^{-10} relative, parameters to 3.7×10^{-7} — at 1.41 – $1.76\times$ the finite-difference speed across the order range (table 2). Correctness is audited independently of speed: fig. 8 compares every column of the analytic Jacobian against a central finite difference on a fitted slice; the worst relative disagreement across all strikes and parameters is 3.7×10^{-6} — of the size of the central difference’s own truncation error at the audit step, so the comparison bounds the analytic route by the reference’s resolution, not by any detected defect.

One honesty note on exactness. Theorem 10.1 is exact for the continuous objects. The implementation interpolates x and G as two separate Hermite interpolants, so between nodes the interpolated price is not algebraically the integral of the interpolated quantile, and the implemented gradient inherits that (small, fourth-order) mismatch. The operational claim is therefore *benchmarked agreement* (fig. 8), not algebraic identity — the same theorem-plus-certificate posture the paper takes everywhere (section 12).

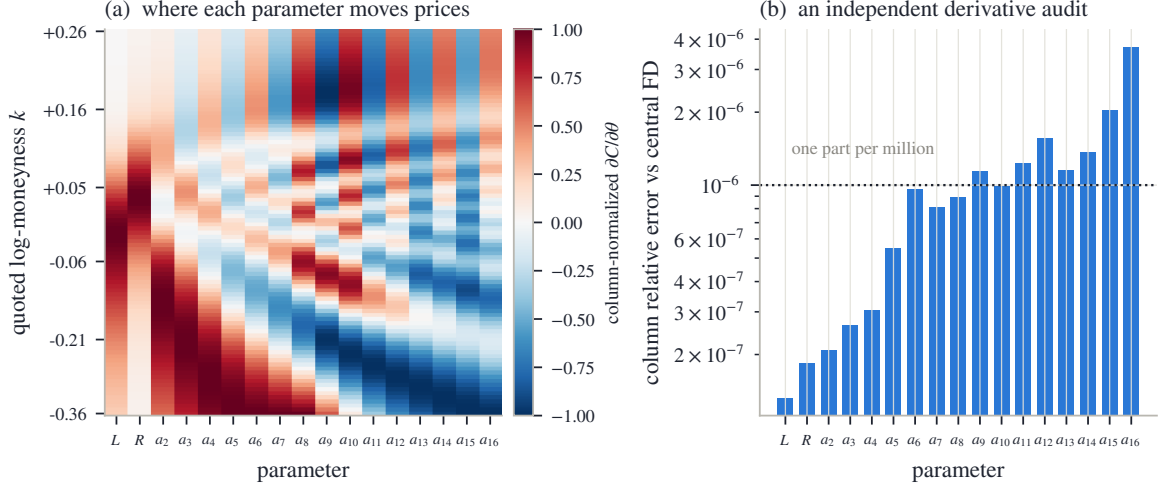


Figure 8: The analytic Jacobian audited against an independent central finite difference, on the fitted SPY December 2026 slice. Panel A: each column is one parameter’s price sensitivity across the strike ladder (columns normalized to $[-1, 1]$): the chord coordinates L and R weight the wings, each Legendre mode paints its own oscillation across the body, and every body column also reaches the wings through the endpoint coupling (16). Panel B: column-wise relative disagreement between the one-pass analytic Jacobian (47)–(49) and the central difference; the worst entry is 3.7×10^{-6} , at the level of the finite difference’s own truncation error — agreement with a derivative computed by a completely different route, to that route’s own resolution.

11 Calendar: order the ledgers

One dimension has been silent: time. Proposition 5.1 silences butterflies within a slice, but a surface is many slices, and fitting them independently leaves a second static arbitrage on the table. The requirement is a comparison of normalized calls at equal *log-moneyness*,

$$c_1(k) \leq c_2(k) \quad \text{for all } k, \quad \tau_1 < \tau_2, \quad (51)$$

and it must be derived, not asserted, because at equal k the two options are struck at *different dollar strikes* ($K_i = F_i e^k$), so (51) is not the plain listed calendar spread it may evoke. The derivation is three lines once the standing assumptions are made to work in the open.

Lemma 11.1 (Calendar ordering at equal log-moneyness). *Assume deterministic rates and carry, so that all forward measures coincide with one risk-neutral measure $\mathbb{P}$, today’s forward curve $t \mapsto F_{0,t}$ is deterministic, and the deflated spot $M_t = S_t/F_{0,t}$ is a positive $\mathbb{P}$ -martingale with $M_0 = 1$. Then $M_{T_i} = S_{T_i}/F_i = Y_i$ is exactly the normalized return of slice i , and for $T_1 < T_2$ and every k ,*

$$c_2(k) = \mathbb{E}[(M_{T_2} - e^k)^+] = \mathbb{E}[\mathbb{E}[(M_{T_2} - e^k)^+ | \mathcal{F}_{T_1}]] \geq \mathbb{E}[(\mathbb{E}[M_{T_2} | \mathcal{F}_{T_1}] - e^k)^+] = \mathbb{E}[(M_{T_1} - e^k)^+] = c_1(k),$$

by the tower property, conditional Jensen applied to the convex payoff, and the martingale property.

The deterministic-carry assumption is doing all the visible work: it is what makes one deflator serve both expiries, hence what makes the *normalized* calls comparable at equal k at all. The monetizing trade behind a violation is correspondingly *forward-scaled* in strike — short $1/(D_1 F_1)$ calls struck at $K_1 = F_1 e^k$ against long $1/(D_2 F_2)$ calls struck at $K_2 = F_2 e^k$ — not the equal-dollar-strike listed spread; the classical statement of the static conditions in

these units is in Carr and Madan [6]. In the language of probability, (51) together with equal means says the laws of Y increase in *convex order*: the longer-dated law is worth at least as much against every convex payoff. Checking a continuum of call inequalities sounds like work. It is not, because the call curve and the ledger are conjugates of each other, and pointwise domination passes through conjugacy in both directions.

11.1 Conjugate duality, and the order theorem

Lemma 11.2 (Ledger–call conjugacy). *For any admissible slice, with $u = \Lambda(z)$ throughout,*

$$c(k) = \max_{z \in [-\infty, \infty]} [G(z) - e^k(1 - \Lambda(z))], \quad G(z) = \min_{k \in [-\infty, \infty]} [c(k) + e^k(1 - \Lambda(z))], \quad (52)$$

the maximum attained at $z = z_k$ and the minimum at $k = x(z)$.

Proof. For any z ,

$$G(z) - e^k(1 - \Lambda(z)) = \int_z^\infty (e^{x(t)} - e^k) \rho(t) dt,$$

whose integrand is negative exactly for $t < z_k$ and positive exactly for $t > z_k$ (monotonicity of x). Starting the integral before z_k adds negative mass; starting it after z_k forgoes positive mass; so the expression is maximal at $z = z_k$, where it equals $c(k)$ by (30). (The endpoints $z = \pm\infty$ give $1 - e^k \leq c(k)$ and $0 \leq c(k)$, consistent.) For the second identity, fix z . By the first identity, for every k ,

$$c(k) + e^k(1 - \Lambda(z)) \geq [G(z) - e^k(1 - \Lambda(z))] + e^k(1 - \Lambda(z)) = G(z),$$

with equality at $k = x(z)$, where the maximizing z' of the first identity is z itself. As $k \rightarrow -\infty$ the expression tends to $\mathbb{E}[Y] = 1 = G(-\infty) \geq G(z)$ and as $k \rightarrow +\infty$ it diverges, so the minimum over the closed line is attained at the finite root. $\square$

The first identity of (52) is the pricing formula (30) rediscovered as a maximization; the pair says the call curve and the ledger are (essentially) Legendre transforms of one another, with the strike root as the point of tangency.

Theorem 11.3 (Calendar order is ledger order). *Let two admissible slices have call curves c_1, c_2 and ledgers G_1, G_2 (both mean one). Then*

$$c_1(k) \leq c_2(k) \text{ for all } k \iff G_1(z) \leq G_2(z) \text{ for all } z.$$

Proof. ($\Leftarrow$) For every k , using the max-representation and $G_1 \leq G_2$ pointwise,

$$c_1(k) = \max_z [G_1(z) - e^k(1 - \Lambda(z))] \leq \max_z [G_2(z) - e^k(1 - \Lambda(z))] = c_2(k).$$

($\Rightarrow$) For every z , using the min-representation and $c_1 \leq c_2$ pointwise,

$$G_1(z) = \min_k [c_1(k) + e^k(1 - \Lambda(z))] \leq \min_k [c_2(k) + e^k(1 - \Lambda(z))] = G_2(z). \quad \square$$

A continuum of option inequalities has become the pointwise ordering of two curves that are already on the grid — the same array that priced the calls and carried the Jacobian, doing its fifth job. Note where the mean-one normalization enters: both ledgers start at $G(-\infty) = 1$, so the ordering is of two curves pinned together at one end, and a legitimate term structure clears the bar by a thin margin everywhere — which is why calendar violations are easy to manufacture and hard to see by eye on smile plots.

Remark 11.4. Pointwise call domination with equal means is the classical characterization of convex order (calls plus affine payoffs generate the convex functions), and by the theorems of Strassen and Kellerer [15, 20] a family of laws increasing in convex order is exactly one that some martingale can run through. Theorem 11.3 is thus the model’s native test for “these two slices could belong to one arbitrage-free market”.

11.2 Enforcement is a control, not a theorem

Within one slice, validity was structural. *Across* slices it is not: the family is closed under nothing that relates two different expiries, so calendar order must be imposed by the objective, and the paper’s obligation is to say exactly what is and is not guaranteed. The production scheme adds, when fitting a far slice against an already-fitted nearer one, hinge rows

$$\sqrt{w_{\text{cal}}} \zeta_m (c_{\text{near}}(k_m) - c_{\text{far}}(k_m))^+, \quad (53)$$

at fixed strikes k_m — with three deliberate scope decisions.

Price space, not ledger space. Theorem 11.3 makes the ledger comparison the natural *measurement*, and it remains the diagnostic of record: rank space needs no volatility inversion and keeps meaning in the far wings, exactly where inversion goes blind. But as a *constraint* the ledger is a trap: $G(z)$ integrates the entire upper tail, so a ledger floor at any single rank drags both slices’ extrapolated wings — curve regions no one quoted — into the constraint. On an acute short-dated pair, a full-grid ledger floor dragged the far expiry’s *worst quote error* from 10.6 to 1095.0 vol bp, a two-order-of-magnitude phantom cost that vanished when the same ordering was read at fixed strikes. The constraint rows (53) therefore compare prices, which are local in strike.

Support confinement, with a smooth taper. The strikes k_m run over the *intersection of the two slices’ retained quote spans* only, and the weights ζ_m taper to zero in a $\cos^2$ ramp (continuously differentiable, so the solver sees no kink) over the outer 15% of that span. Outside the common quoted support the constraint is silent by design: enforcing order between two extrapolations would let one model prior push another around in a region where the market said nothing. The count of constraint strikes scales with basis order ($\max(49, 4N + 1)$ nodes), so a high-order slice cannot slip a violation between nodes of a fixed coarse grid.

Finite weight, honest reporting. The rows carry a large but finite weight; a hard-pressed fit can retain a small violation rather than destroy its quote fit. Residual violations are measured and reported in economic units — worst-violation strike, normalized-price gap, and the gap as a fraction of the local bid–ask width — so that “violation of 10^{-6} ” resolves into “a fortieth of the spread: unharvestable” or into a number that stops publication.

The guarantee, stated plainly: *within the common quoted support, calendar order holds up to a reported, economically sized tolerance; outside it, calendar order is not enforced, and the ledger diagnostic reports whatever the priors did.* This is a control, not a theorem, and the distinction is visible in live data.

A live illustration. In the frozen snapshot (appendix C), the two shortest SPY expiries — two and four *calendar* days from the reference date — are fitted cleanly, each valid on its own, calendar rows active on their common support. Yet the pair trips a *vol-space* calendar audit run at equal log-moneyness over the full display grid, both legs’ total variances read as volatilities on the far node’s clock: a crossing of 1840.6 vol bp, argmax at $k = +0.981$. The two slices’ common quote span is $[-0.028, +0.016]$: the flagged strike sits dozens of span-widths to the right of anything quoted, where the two call prices are 5.4×10^{-107} (two-day leg) and 1.6×10^{-26} (four-day leg) of the forward. A price-space audit over the same grid finds a worst violation of 2.2×10^{-15} of forward — round-off, not economics. What the vol-space audit detected is real — the two *extrapolations* cross, in a region where implied variance is astronomically sensitive to price — and harvesting it would mean trading options dozens of span-widths out of the money at those prices: there is no arbitrage there, only two priors disagreeing about a wilderness. Figure 9 draws the pair, the common span, and the crossing; the paper’s position is that a fitter should enforce where the market speaks, report honestly where it does not, and resist the temptation to make the wilderness look tidy by constraining it.

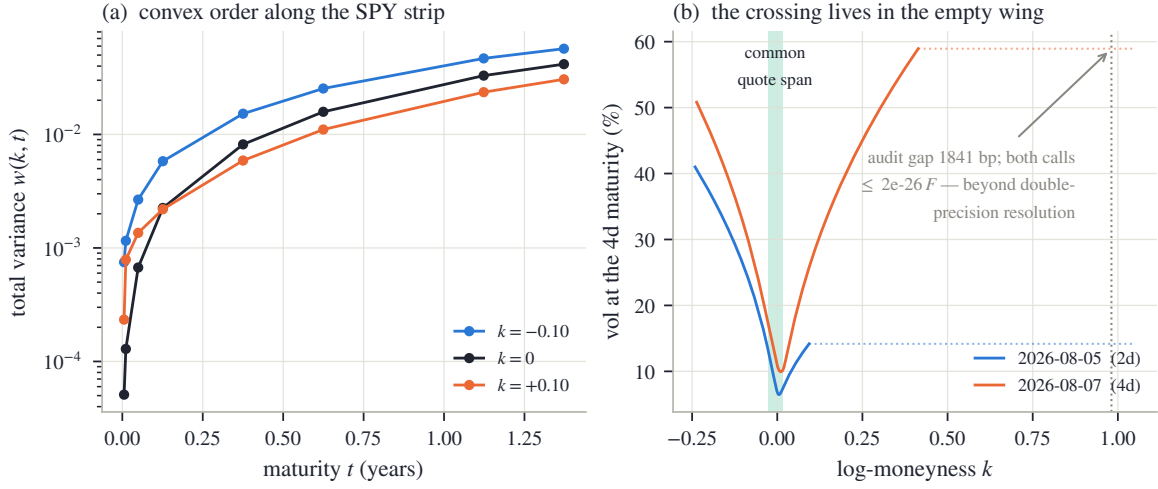


Figure 9: Calendar order as control, in two panels. Panel A: SPY total-variance term structure at several fixed log-moneyness values from the frozen snapshot — monotone through the quoted region, as the constraint rows (53) require. Panel B: the confinement illustration on the two- and four-calendar-day pair, both curves drawn as $\sqrt{w(k)/\tau_{\text{far}}}$ — total variances annualized on the common far-node clock, so the vertical axis reads in volatility units and the audit gap is the visible gap between the curves. The common quote span is shaded, and within it the curves are ordered; each solid curve is truncated where its node’s call price drops below 10^{-13} of forward and continued as a light dotted flat — beyond those points the stored smile is Black-inversion noise, which is itself the argument for support-confined enforcement. The audit crossing at $k = +0.981$ (gap 1840.6 vol bp, call prices 5.4×10^{-107} and 1.6×10^{-26} of forward, annotated) lies in that censored region: a disagreement between priors in an economically empty region, reported, not enforced away.

12 Computation: earning the theorems

The propositions of section 5 are statements about continuous objects. An implementation prices through quadrature and interpolation, and an interpolant inherits nothing automatically. This section gives the numerical analysis in full: the quadrature, the optimization charts, a catalogue of the floating-point failure modes and their remedies, and — the section’s organizing principle — the *certificates* by which the discretization earns, rather than borrows, the continuous theorems.

12.1 The quadrature

All integrals live on a fixed symmetric grid $z_j \in [-Z_{\max}, Z_{\max}]$ in log-odds, with $Z_{\max} = 40$ and 8001 points (odd, so $z = 0$ is a node); optimization iterates on a coarser 2001-point grid and the accepted fit is rebuilt once on the full grid (converged parameters agree to 1.4×10^{-6} between the two). One build is five steps: evaluate g at the cached ranks (one matrix–vector product — g is linear in θ , and the Legendre matrix is cached); integrate e^g cumulatively (fourth-order Simpson) into $\bar{x}$, anchored at the median node; normalize, $m = -\log I$; integrate $e^x \rho$ cumulatively from the right into G ; stop. Nothing is iterative, and the exact nodal derivatives $x' = e^{g \circ \Lambda}$ and $G' = -e^x \rho$ come free, so off-grid evaluation uses cubic *Hermite* interpolation — locally $O(h^4)$ with no spline solve — and the strike root $x(z_k) = k$ is polished by Newton on the interpolant to machine precision, which is what makes the ATM handles of proposition 8.1 exact rather than grid-limited.

Truncation at $|z| = Z_{\max}$ is not prayed away but summed: beyond the grid the integrands are exponentials to $O(e^{-Z_{\max}})$ relative accuracy, so the build appends the closed-form tail corrections

$$\int_{Z_{\max}}^{\infty} e^{\bar{x}} \rho \, dz \approx \frac{e^{\bar{x}(Z_{\max}) - Z_{\max}}}{1 - \lambda_+}, \quad \int_{-\infty}^{-Z_{\max}} e^{\bar{x}} \rho \, dz \approx \frac{e^{\bar{x}(-Z_{\max}) - Z_{\max}}}{1 + \lambda_-}, \quad (54)$$

(the right-hand denominator displaying the wall once more). Strikes beyond the grid’s return range are priced on the slice’s own tail asymptote, and the formula is worth deriving in the two lines it takes, because its $\lambda_+/(1 - \lambda_+)$ factor looks mysterious until the subtraction is seen. For k past the grid edge, the strike root extrapolates linearly along the tail, $z_R(k) = Z_{\max} + (k - x(Z_{\max}))/\lambda_+$; on that tail the ledger entry and the cash leg of (30) are both single exponentials,

$$G(z_k) \approx \frac{e^{k - z_R(k)}}{1 - \lambda_+} \quad (\text{the tail correction (54) restarted at } z_R(k), \text{ where } x = k), \quad e^k(1 - u_k) \approx e^{k - z_R(k)},$$

and their difference gives

$$c(k) = e^{k - z_R(k)} \left(\frac{1}{1 - \lambda_+} - 1 \right) = e^{k - z_R(k)} \frac{\lambda_+}{1 - \lambda_+} \quad (55)$$

(mirrored on the left for puts): continuous at the seam with (54) and Lee-consistent by construction. The asymptote is not decoration — the display edge of a short-dated smile genuinely reaches it.

12.2 Three charts, one optimum

The optimizer may run in any of three coordinate systems on the same family; the objective is identical, so the optimum is chart-independent (verified numerically on live nodes: parameters agree across charts to 2.2×10^{-7}), and the choice is purely about conditioning and constraint geometry.

Raw: $\theta = (L, R, a_2, \dots, a_N)$. Simple, but body and tails are coupled (16), and the wall $\lambda_+ < 1$ is a boundary the trust region must respect.

Endpoint: $\phi = (\log \lambda_-, \log \lambda_+, a_2, \dots, a_N)$, with $\theta = M\phi$ the exact unit-determinant linear back-solve of (16). Equivalently, the body basis becomes the endpoint-vanishing combinations $P_n(1 - 2u) - (1 - u) - (-1)^n u$: same polynomial space, different names on the axes. Body moves now hold both tail scales fixed *by construction* — the chart in which tail priors, manual edits, and warm starts speak the language they mean.

Logistic (production default): additionally set $\lambda_+ = \text{expit}(\xi)$, $\xi \in \mathbb{R}$. Every point of $\mathbb{R}^{N+1}$ is now admissible: the wall is not respected but *unreachable*, and the word “unconstrained” in section 5 becomes a literal statement about the production solve. Near the wall the chart compresses —

$$\frac{d \log \lambda_+}{d\xi} = 1 - \lambda_+, \quad (56)$$

so trust-region steps shrink, in $\log \lambda_+$ terms, exactly where the raw chart needed a barrier; far from the wall $\xi \approx \log \lambda_+$ and nothing changes. One honest footnote: the logistic chart removes the mathematical wall, not the floating-point one — beyond $\xi \approx 36.7$, $\text{expit}(\xi)$ rounds to exactly 1 — so the build retains a hard admissibility guard $\lambda_+ \leq 1 - 10^{-6}$ regardless of chart.

12.3 The floating-point catalogue

Structural theorems do not survive IEEE-754 by themselves. The catalogue below is the complete list of the failure modes met in practice, each with its remedy; every one of them was found either by the randomized audit described next or by a production incident, and each is now locked by a regression test. (General background: 13.)

1. *Rank saturation.* For $|z| \gtrsim 36.7$ (i.e. $e^{-|z|} < 2^{-53}$), $\text{expit}(z)$ rounds to exactly 1, so the factor $1 - u$ silently becomes 0 and $u(1 - u)$ becomes 0. Remedy: never form $1 - u$ from u ; use $u(1 - u) = \text{expit}(z) \text{expit}(-z)$ (each factor evaluated on its own accurate side) and keep cash legs in log space, $e^k(1 - u_k) = \exp(k - \log(1 + e^{z_k}))$ via a stable log-sum-exp. This failure was live once: the cash leg of (30) collapsed to zero in the far right wing and the priced call stepped up to the bare ledger $G(z_k)$; the audit caught it as a butterfly violation of order 1.9×10^{-2} , and the log-space remedy eliminates it — the re-audit at the same strikes and stencil finds a worst violation of exactly 0.
2. *Interior overflow.* Endpoint admissibility does not bound the body: cancellation in (16) can keep $\lambda_{\pm}$ tame while an interior excursion of g overflows e^g or the running integral. Remedy: an explicit exponent budget — the build refuses, with a clean exception the calibrator’s penalty branch absorbs, any trial whose g or unnormalized transport exceeds 700 (just under log of the double-precision ceiling, ≈ 709.78). A guard at the wall cannot substitute: it watches the endpoints, and this failure lives in the middle.
3. *The Lee map at large moment order.* The textbook form $\Psi(r) = 2 - 4(\sqrt{r^2 + r} - r)$ subtracts two nearly equal large numbers and, for $r \gtrsim 10^{15}$, returns ≈ 2 where the truth is ≈ 0 . Remedy: rationalize once, $\sqrt{r^2 + r} - r = r(\sqrt{1 + 1/r} - 1) = 1/(\sqrt{1 + 1/r} + 1)$, giving the algebraically identical

$$\Psi(r) = 2 - \frac{4}{\sqrt{1 + 1/r} + 1}, \quad (57)$$

which subtracts nothing large and is accurate over the whole range. Companion guards: $\lambda \rightarrow 0$ maps to $\beta = 0$ (the finite limit, not a division error), and $\lambda_+ \geq 1$ reports the ceiling $\beta_R = 2$.

4. *Degenerate quadrature inputs.* The parameter-independent grid arrays are cached and returned read-only, so an accidental in-place write errors instead of corrupting every later build — a hygiene rule, recorded because the failure it prevents is silent and global.

12.4 Certificates: the discretization earns the theorem

proposition 5.1 is about the continuous slice. The implementation prices through two Hermite interpolants, and ordinary cubic Hermite pieces are *not* monotonicity-preserving in general: positive nodal derivatives do not preclude segment overshoot. The honest guarantee is therefore a pair: a structural theorem for the continuous object, plus a *certificate* and a measured tolerance for the implementation. Both are cheap; both run in production; neither is taken on faith.

The monotonicity certificate. For a cubic Hermite interpolant of an increasing function on a grid, the Fritsch–Carlson condition [9] is sufficient: on each segment, monotonicity holds if both endpoint derivatives lie in $[0, 3 \times \text{secant slope}]$. With exact nodal derivatives of a smooth curve the derivative-to-secant ratio is $1 + O(h^2)$, far inside the region, so the scan passes with margin — and the *margin* (the worst distance to the boundary of the Fritsch–Carlson box, over all active segments) is the certified quantity, reported per slice. Segments where the ledger has decayed below round-off are certified flat-to-tolerance rather than monotone; the caveat is part of the certificate, not a footnote to it.

The belly certificate. Convexity of the priced smile over the traded range is re-checked from the model’s own derivatives: the function g_D of (5), evaluated at 801 points across the quoted strike range from analytic w, w', w'' (never from differenced prices), must clear -10^{-4} . The threshold has stated units because (5) is an exact factorization: g_D is normalized so a flat smile gives $g_D \equiv 1$, and a reading of -10^{-4} tolerates a density no more negative than 10^{-4} of the local Black density factor $\varphi(d_-)/\sqrt{w}$ — a relative tolerance, not an unexplained constant. The grid is dense because the classical counterexamples are narrow — a known arbitrageable SVI parameter set dips to only -0.033 over a width of ≈ 0.02 in k — and the check costs 1.5 ms, so it runs on every publish and a failure blocks it.

The randomized audit. What certificates do not cover — convexity at finite butterfly widths, digital bounds, grid adequacy — a battery measures. 27 parameter vectors are drawn through the logistic chart (hence all admissible by construction) with a fixed seed, stratified across orders 4, 8, and 16 and three stress classes — plain, near-wall, and wild-body — with equal draws per cell. Each slice is priced at randomized sub-grid strikes and audited *in the strike variable* $y = e^k$ — convexity lives there, per proposition 5.1(iv), and checking it in k was the battery’s own first bug. Worst violations across the battery: call-price bounds 1.1×10^{-11} , butterflies 8.2×10^{-13} , digital bounds 1.5×10^{-12} , and agreement with a $4\times$ finer reference quadrature at random sub-grid strikes 2.3×10^{-9} — one tolerance set, uniform across the classes. The quotable summary: *the implementation’s residual arbitrage is bounded by the worst figures just quoted, in normalized price, across 27 adversarially drawn slices* — which is the sense in which “cannot have arbitrage” becomes “measured not to, at tolerance so-and-so”.

12.5 Timing

Table 2 reports wall-clock medians and interquartile ranges for one-slice calibrations of a 40-quote synthetic strip on the reference implementation (single-threaded, warm caches), from one interleaved run alternating analytic and finite-difference Jacobians — never best-of- n . Two live-data numbers orient the reader alongside the table: the 94-quote deep-dive SPY node of section 13 fits warm in about 42 ms at the shipped order $N = 16$, and the 17-quote shortest NVDA node at its guarded order in about 24 ms. Wall-clock ratios are machine- and cache-dependent; the structural claim is only the $O(P n_{\text{grid}})$ accounting of section 10.

Configuration	Order	Median	IQR	vs. FD Jacobian
Analytic Jacobian	6	15.9 ms	0.5 ms	$1.41\times$
Finite difference	6	22.5 ms	2.7 ms	—
Analytic Jacobian	12	26.0 ms	2.1 ms	$1.68\times$
Finite difference	12	43.8 ms	3.2 ms	—
Analytic Jacobian	16	28.9 ms	0.5 ms	$1.76\times$
Finite difference	16	50.9 ms	3.9 ms	—

Table 2: Single-slice calibration timings: median and interquartile range from one interleaved run (alternating analytic and finite-difference Jacobian solves, after warm-up). The analytic column ratio is the median finite-difference time divided by the median analytic time for the same configuration; measured range 1.41–1.76 $\times$.

13 Examples

All real-market examples in the paper are fitted from one frozen snapshot: SPY and NVDA listed options, eight expiries each, from 2 calendar days to 1.3726 years, captured after the close of the 2026-08-03 US session from a delayed consolidated feed, with spots 757.67 (SPY) and 206.80 (NVDA). Fits use the haircut target (44), the logistic chart, the shipped order $N = 16$ under the guard (45), and the calendar rows (53); nothing was tuned per example. One consequence of the recipe should be recalled from section 9 before reading the figures: SPY spreads are all tighter than the haircut width, so every SPY band degenerates to a mid target and the SPY panels draw raw bid–ask quotes, while the NVDA nodes carry live bands. Across all 16 fitted nodes the median quote-fit error is 5.9 vol bp and the worst 24.6 vol bp (the shortest NVDA node, discussed below). Provenance, quality counts, and regeneration instructions are in appendix C; the synthetic examples of this section are exactly specified there as well.

13.1 The SPY gallery

Figure 10 shows all eight SPY expiries: quotes as raw bid–ask whiskers with their mids, fitted slices through them, per-expiry error annotated. The gallery’s purpose is not to impress with residuals — a flexible basis fitting tight liquid mids *should* produce small residuals — but to exhibit the shape vocabulary of one underlier across maturities: the steep short-dated put skew relaxing into the long-dated smile, the at-the-money level drifting with the term structure, the wings beyond the outer quotes following the two-parameter exponential-tail prior of section 6. Median error across the eight slices is 3.6 vol bp, worst 7.5.

13.2 One node under the microscope

Figure 11 takes the December 2026 SPY expiry — the slice already used in figs. 4, 5 and 8 — and reads it three ways: fit against the raw bid–ask quotes, residual ladder in vol bp per strike, and the fitted density against an ATM-matched normal. The node fits 94 quotes to 3.3 vol bp rms. Its exact handles (proposition 8.1): ATM volatility 14.76%, skew $\sigma'(0) = -0.430$, curvature $\sigma''(0) = 4.40$. The fitted forward rank is $u_* = 41.31\%$ — below one half, and, what actually carries the skew, below the matched flat-Black percentile $\Phi(d)$: this is the digital mismatch $\Delta < 0$ of proposition 8.1 in live numbers, one sign with the measured $\sigma'(0) < 0$, and the contrast with the symmetric slice of section 7 (whose forward rank sits *above* one half with $\Delta \approx 0$) shows which effect owns which comparison: the martingale shift moves u_* against one half, the skew moves it against $\Phi(d)$, and the second comparison is the one the smile trades on. The var-swap strike (35) is 19.84%, well above the ATM volatility — as expected for a slice

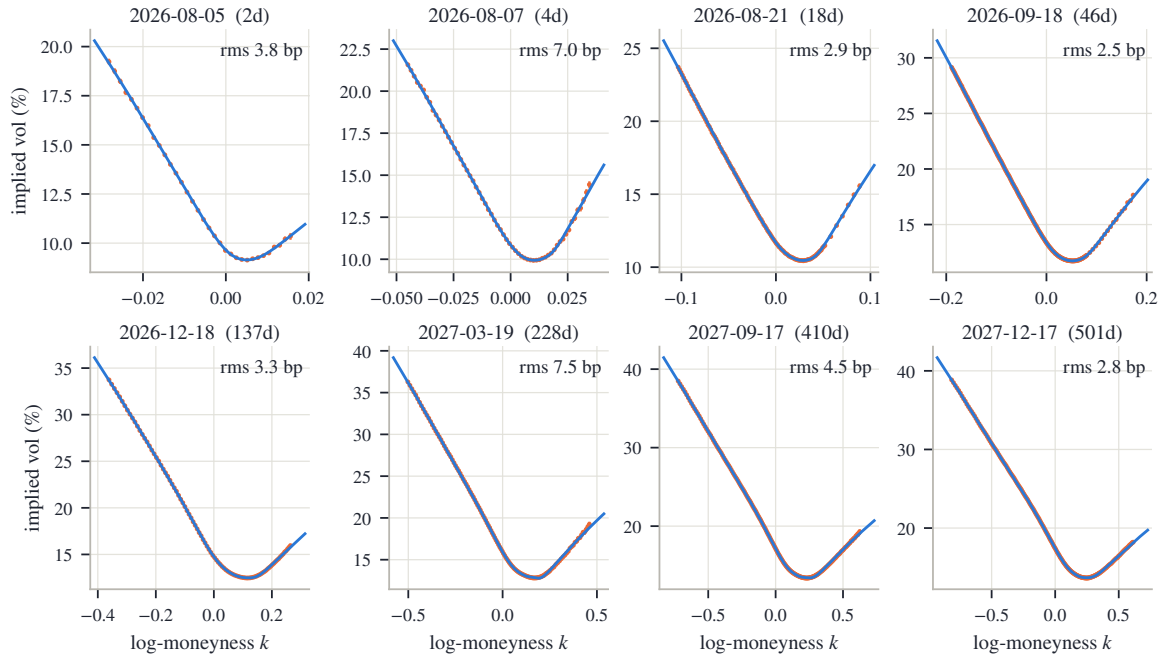


Figure 10: Eight SPY expiries from the frozen snapshot, fitted with one recipe. Each panel: raw bid–ask quotes (whiskers) and mids, fitted LQD slice (line), per-expiry rms in vol bp (annotation). The haircut bands do not appear because none survive on SPY — every spread is tighter than $2h$, so the fit target is the mid everywhere (section 9). Quoted strike coverage shrinks and steepens toward the short end; everything beyond the outermost quotes in each panel is the model’s exponential-tail extrapolation, labeled as prior, not data, per section 6.

with this much left tail, since the log contract weights exactly the wings the smile lifts. The density panel shows where that premium lives: mass moved from the right shoulder into the left tail relative to the matched normal.

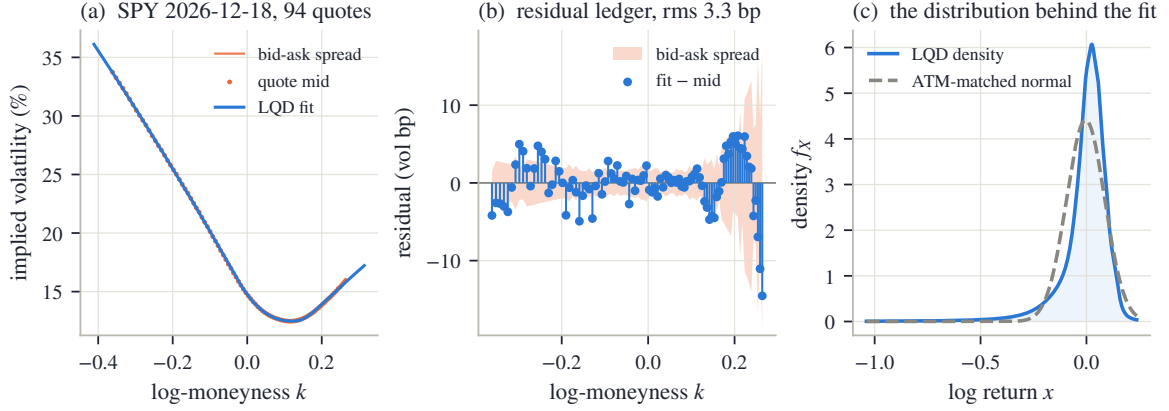


Figure 11: The SPY December 2026 node. Panel A: raw bid–ask quotes and the fitted slice (no haircut bands survive on SPY — section 9). Panel B: the residual ladder — signed fit error versus mid in vol bp at each quote, drawn inside the bid–ask envelope; rms 3.3 vol bp. Panel C: the fitted density against a normal matched at the money: the skew is a transfer of mass into the left tail, paid for by a thinner right shoulder — the same fact the negative digital mismatch $\Delta < 0$ of proposition 8.1 states in one number.

A ticket priced by hand. The ledger makes a price auditable with a pencil, which is worth doing once in public, on a contract that actually trades. On this node, price the listed December 800 call: log-moneyness $k = 0.0409$ against the node’s forward. The strike root gives rank $u_k = 65.27\%$: the option finishes in the money with probability $1 - u_k$. The two ledger entries (30) read $G(z_k) = 0.378366$ — the share of the forward carried by outcomes above the strike — and $e^k(1 - u_k) = 0.361834$ — the strike bill paid on that event — so

$$c(0.0409) = 0.378366 - 0.361834 = 0.016532,$$

or \$12.70 per share in dollars through the node’s forward and discount, for an implied volatility of 13.37%. Reproducing these five numbers from the snapshot file is a one-screen exercise and the fastest full-stack audit of an implementation the author knows.

13.3 NVDA, including the shortest node

Single-name smiles are heavier-tailed and coarser-quoted — and, per section 9, NVDA is where the haircut band mechanics actually engage, with roughly half the quotes carrying live bands on the featured nodes. Figure 12 shows two NVDA expiries: the shortest node (2 calendar days from the reference date) and the longest, December 2027. The long node behaves like SPY’s slices with wider tails (fig. 6): 69 quotes, rms 7.4 vol bp, fitted at full order. The shortest node is the order guard (45) working in public: 17 retained quotes admit effective order $N_{\text{eff}} = 7$, and the fit lands at 24.6 vol bp rms — the largest error in the snapshot, on the thinnest, jumpiest strip, with the fewest parameters: exactly where a fitter should be least confident, and visibly so in the figure’s wider bands.

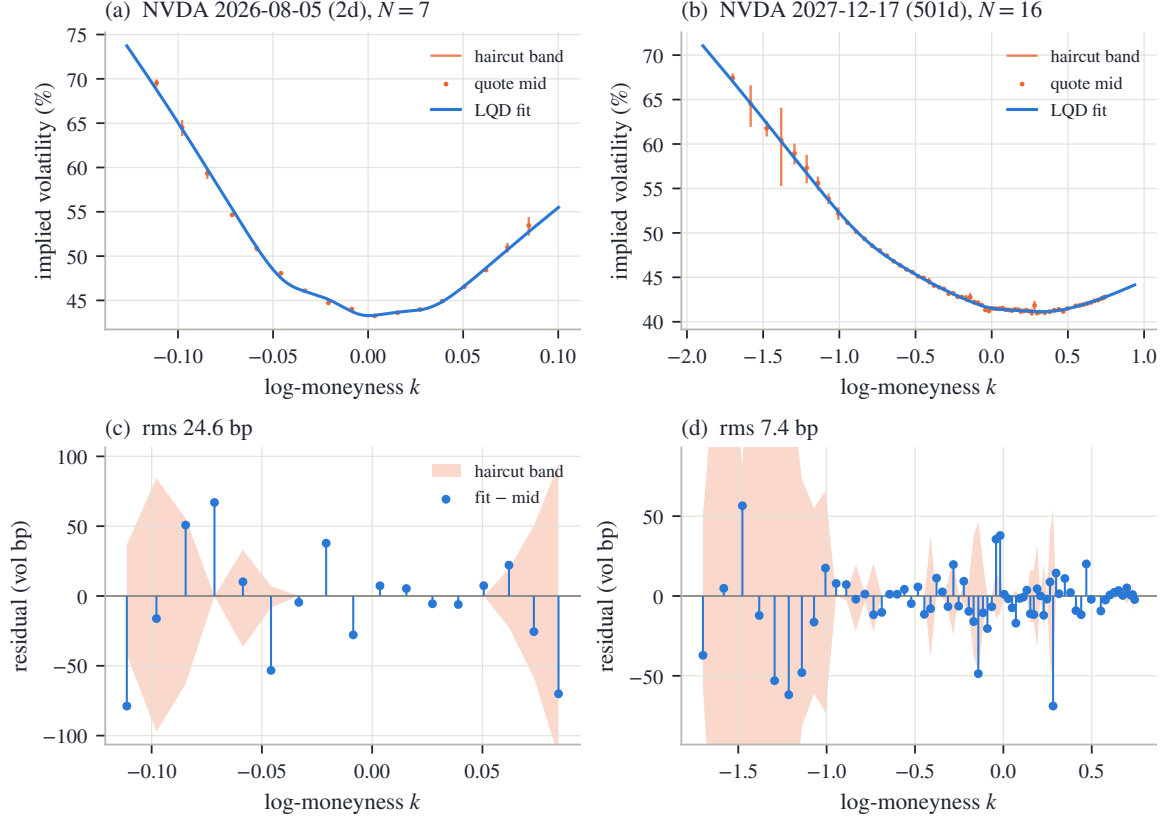


Figure 12: Two NVDA nodes from the same recipe, with the haircut bands live (unlike SPY's, NVDA's spreads are wide enough to survive the shrink (44)). Left: the shortest expiry (2 calendar days) — 17 quotes, order-guarded to $N_{\text{eff}} = 7$, rms 24.6 vol bp; the guard trades inter-quote flexibility for stability on a strip that cannot support sixteen modes (section 9.4). Right: the December 2027 expiry at full order — 69 quotes, rms 7.4 vol bp — with the single-name tail scales visibly heavier than SPY's (fig. 6).

13.4 Order control: the synthetic double-hump

What does basis order buy? Section 5 said the family can approximate bimodal laws; this example measures it. The target is a known mixture of two lognormal components (parameters in appendix C), martingale-normalized, quoted as noiseless implied volatilities on a realistic strike grid — an approximation benchmark, not a market fit. At $N = 16$ the fit reproduces the target smile to 4.8 vol bp rms (maximum error 11.32 vol bp) and recovers the bimodal density *faithfully*: modes at -0.099 and 0.088 against the true -0.101 and 0.089, trough and asymmetry intact (fig. 13). At $N = 6$ (fig. 14) the smile moves little — the two fits agree in implied volatility to 41.8 vol bp on a dense grid, with the larger gaps sitting out in the wings beyond the quoted ladder — and the density keeps both modes (2, at this mixture separation) but is *distorted*: the peaks overshoot, the trough is half-filled, the mode locations drift. The honest scalar carrier of the difference is the L^1 distance to the true density, which more than doubles: 0.047 at $N = 16$ against 0.108 at $N = 6$.

The pair of figures is the paper’s cleanest exhibit on information versus convention. The *quotes* barely distinguish the two fits; the *densities* differ measurably — not, at this separation, by the existence of the modes, but by their fidelity: shape, depth, and placement of the structure between the quotes. Below the quote spacing, that fidelity is chosen by the basis order, not by the market — so a desk that trades density-sensitive structures (digitals, tight butterflies) around events should set N as a conscious convention, and a desk that only reads smiles may never notice the difference. Order is power to assert structure between quotes; the guard of section 9.4 caps that power when quotes are few.

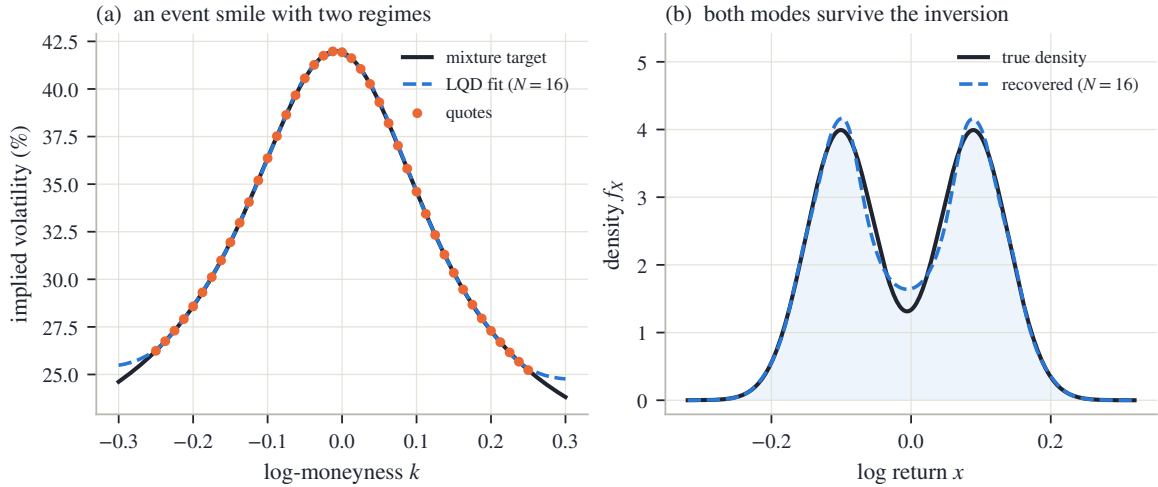


Figure 13: The double-hump benchmark at $N = 16$. Panel A: target implied volatilities from the bimodal mixture and the fitted slice — agreement to 4.8 vol bp rms; note the smile shows bimodality only as gentle shoulder structure. Panel B: the true mixture density (shaded) and the fitted LQD density (dashed): both modes, the trough, and the asymmetry are recovered, with fitted mode locations -0.099 / 0.088 against the true -0.101 / 0.089; positivity was never at risk while the high modes carved the trough.

13.5 The exact audit

The fifth example is the oldest kind: against Euler. Figure 3 (section 4.3) already confronted the implementation with the constant-speed slice’s closed forms across the admissible scale range — transport map to 1.1×10^{-11} , shift to 3.0×10^{-9} (growing lawfully toward the wall), martingale check to 3.3×10^{-16} . The example is listed again here because of what it covers:

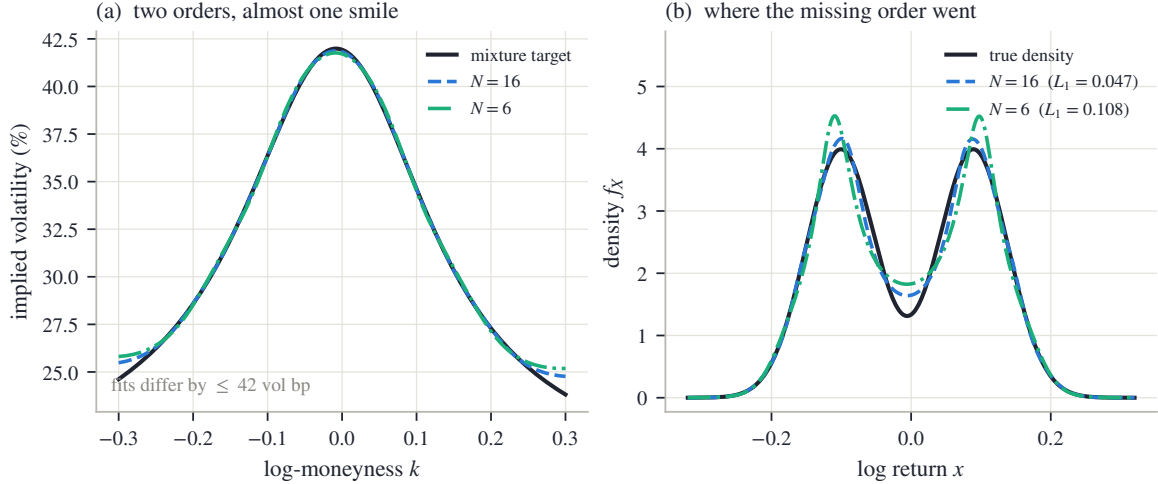


Figure 14: The same target at $N = 6$. Panel A: the $N = 6$ and $N = 16$ smiles nearly coincide (maximum gap 41.8 vol bp, the larger gaps beyond the quoted ladder) — the quotes can barely tell the fits apart. Panel B: the densities can: the $N = 6$ fit keeps both modes but overshoots the peaks, half-fills the trough, and drifts the mode locations — L^1 error to the truth 0.108 against 0.047 at $N = 16$, more than double. Between the quotes, order is a convention, and this is what choosing it asserts.

quadrature, tail corrections, normalization, interpolation, and root-solving in one line of algebra with no free parameters. A wrong grid constant, a mis-anchored cumulative sum, or a stale cache moves any of these agreements by many orders of magnitude; it is the cheapest regression test the model owns, and the recommended first test of any reimplemention from this paper.

14 Related work

The ingredients of this paper are classical, and it is worth saying precisely which shelf each came from and what, if anything, is new in the assembly.

Densities from option prices. That the call curve’s second strike derivative is the risk-neutral density is Breeden and Litzenberger’s [4]; the static no-arbitrage characterization of call quotes used in definition 2.1 is classical and is stated compactly by Carr and Madan [6]. This paper’s (34) is the same identity read in the model’s coordinates, where it costs one division.

Wing asymptotics. The moment formula bounding wing slopes is Lee’s [16]; the upgrade from lim sup to limits under regular variation is Benaïm and Friz’s [1]. The contribution here is only the closed composition (26) — the model’s tail coordinates happen to make Lee’s formula an explicit two-parameter map — and the effective-slope honesty figure.

Implied-volatility parametrizations. SVI [10] and its surface extension SSVI [11] are the industry benchmarks for the volatility chart, with explicitly characterized (but parameter-coupled) no-arbitrage regions; Fong’s arbitrage-free smoothing [8] represents the spline-with-constraints tradition in the price chart. Both approaches place validity in the optimizer; the present model places it in the coordinates. The trade is not free — SVI’s five parameters are quotable desk vocabulary in a way sixteen Legendre coefficients are not — and section 8 plus the package machinery of appendix A are the model’s answer, not a dismissal of the need.

Quantile-based modeling. Modeling distributions through quantile functions goes back at least to Parzen [18] and is developed monographically by Gilchrist [12]; Keelin’s metalog family [14] builds flexible unbounded distributions as polynomial expansions of the quantile function itself and is the closest relative in the applied literature. The two design differences

are load-bearing: LQD expands the *logarithm* of the quantile density — so monotonicity is structural at every order, where the metalog must verify feasibility of its coefficient vector — and it carries the $-\log u - \log(1 - u)$ skeleton explicitly, so the expansion coefficients control tails through two endpoint values with the martingale condition solvable by one shift. In functional data analysis, Petersen and Müller [19] use the same log-quantile-density transform to map density-valued data to an unconstrained Hilbert space — the identical unconstrainedness argument, made for statistical rather than arbitrage reasons; this paper adds the financial boundary conditions (martingale normalization, the $\lambda_+ < 1$ wall, Lee consistency) and the pricing calculus.

Variance swaps and the log contract. The log-contract replication is Neuberger’s [17], its dealer-standard exposition Demeterfi, Derman, Kamal, and Zou’s [7], and the general payoff-replication frame Carr and Madan’s [5]. Here the fitted law is explicit, so (35) integrates the density directly rather than replicating across a strip.

Calendar order. That call-price domination with equal means is convex order, and that convex-ordered marginals are exactly those a martingale can traverse, is Strassen’s and Kellerer’s [15, 20]. Theorem 11.3 restates the first fact as a pointwise comparison of the model’s native computational object; the conjugate-duality proof is elementary and surely folklore in some notation, but the author has not found the ledger form stated.

Numerics. Monotone cubic interpolation certificates are Fritsch and Carlson’s [9]; floating-point discipline follows Higham [13]; the solver is the trust-region reflective method [3] as implemented in SciPy [22].

What, then, is claimed as new? Not the transform, not the moment formula, not the conjugacy of calls and tail expectations. The claims are architectural: the separation of validity, information, and convention into structure, audit, and disclosed choices respectively; the single-ledger organization in which pricing, differentiation (theorem 10.1), density, variance swap, and calendar order are five reads of one array; the closed ATM microscope (38) with the skew-as-digital-mismatch reading; and the certificate-plus-audit doctrine of section 12 by which the discretization earns the continuous theorems. Readers are invited to judge the assembly, since every part has prior art somewhere.

15 Limitations, and the contract

A model is used safely in proportion to how precisely its limits are stated. This closing section collects everything the paper does *not* claim, most of it already flagged in place, and ends with the contract table.

What the family cannot represent. Per proposition 5.3: no atoms (a pinned settlement print, a tender-offer spike), no bounded support, no interior zero-density gaps, no default mass at zero, no Gaussian or faster tails. All of these are *smoothed*, not represented; the double-hump example shows the smoothing is graceful and order-controlled, but graceful smoothing of a genuine atom is still a misstatement of the law, and a desk pricing digitals across a known pin should know it.

Tails are priors. The two tail coordinates are weakly identified by any realistic strip (section 6). The model’s virtue is not that it extracts tail information others miss — it is that its tail convention is two readable numbers, Lee-consistent by construction, overridable by policy, and honestly separable from what the quotes pin. Reported values of $\lambda_{\pm}$ and $\beta_{\pm}$ should travel with the fitting recipe that selected them.

Calendar enforcement is a control. Theorem 11.3 is exact for the continuous slices; production enforcement is finite-grid, finite-weight, and confined to the common quoted support (section 11.2). Outside that support, ordering between extrapolations is reported, not enforced; the illustration on the two shortest SPY expiries is the standing example of why this is the

right scope and of what the diagnostic then shows.

The continuous theorems are earned, not inherited. Every guarantee in section 5 passes to the implementation only through the certificates and the audit of section 12, whose worst measured violations are the real numbers a risk committee should file. A reimplementaion from this paper inherits the theorems and none of the certificates until it runs them.

Single-expiry statics only. The model prices one marginal law per expiry. It says nothing about dynamics — no spot-vol correlation process, no forward smile, no jump identification — and the var-swap identity (35) inherits the log contract’s idealizations. Coupling to dynamics, and joint fits across underliers, are outside the paper.

Performance claims are scoped. Timings are medians with IQRs from one machine and one interleaved protocol (table 2); the portable claim is the complexity accounting, not the milliseconds.

Promised (structural)	Controlled (audited)	Not claimed
Positive density; unit mean; butterfly-freedom per slice (proposition 5.1)	Implementation validity at measured tolerance (section 12.4)	Representation of atoms, gaps, bounded support, super-exponential tails
Exponential tails with readable scales; exact moment strip; Lee consistency (propositions 6.2 and 6.3)	Calendar order on the common quoted support, reported in economic units (section 11.2)	Tail values as market information; wing quotes from $\beta_{\pm}$
Exact ATM level, skew, curvature (proposition 8.1); exact analytic Jacobian (theorem 10.1)	Fit quality per node; identifiability via order guard and multi-start (section 9)	Dynamics; realized-variance equivalence; cross-expiry structure beyond convex order

Table 3: The contract: what is guaranteed by construction, what is enforced and measured, and what is explicitly not claimed.

The design thesis, restated once in closing. Validity, information, and convention are different kinds of statement, and a fitter’s architecture should keep them different: validity belongs in the coordinates, where it cannot be traded away by an optimizer under pressure; information belongs in audits, where it is measured rather than presumed; convention belongs in daylight, where it can be argued about. The LQD construction is one demonstration that the separation is achievable at production cost — one array, five uses — for the class of smiles that equity markets actually quote.

A Implementation

This annex records the reference implementation in enough detail to reproduce it: the build pipeline, call evaluation, the analytic sensitivity pass, and the monotonicity certificate, as lightly adapted NumPy-style code in the paper’s notation, followed by the defaults table and practical recommendations. The snippets favor clarity over micro-optimization but preserve every numerically load-bearing line.

A.1 The slice build

Listing 1: One slice build: $\theta \rightarrow (x, x', G, G')$ on the fixed grid. Comments give the paper’s equation numbers.

```
# Cached, parameter-independent (read-only): grid z, ranks u = expit
# (z),
# weight rho = expit(z)*expit(-z) # never u*(1-u): rank saturation
# and the Legendre matrix Pmat[n, j] = P_n(1 - 2 u_j).
```

```

def build(theta, grid):
    L, R, a = theta.L, theta.R, theta.a
    g = (1 - grid.u)*L + grid.u*R + a @ grid.Pmat          # eq. (lqd)
    if g.max() > 700.0 or not np.isfinite(g).all():
        raise InteriorOverflow                               #
        catalogue #2
    dx = np.exp(g)                                           #  $x'(z)$ ,
        eq. (speed)
    xbar = cumulative_simpson(dx, dx=grid.dz, anchor=grid.mid)
    if xbar.max() > 700.0:
        raise InteriorOverflow
    lam_m, lam_p = np.exp(g[0]), np.exp(g[-1])              # eq. (
        speeds)
    if lam_p >= 1.0 - 1e-6:
        raise InadmissibleTail                              # the wall
    mass = np.exp(xbar) * grid.rho
    I = trapz(mass, dx=grid.dz) \
        + np.exp(xbar[-1] - grid.z_max) / (1.0 - lam_p) \
        + np.exp(xbar[0] - grid.z_max) / (1.0 + lam_m)      # eq.
        (tailcorr)
    m = -np.log(I)                                           # prop. (
        shift)
    x = m + xbar
    dG = -np.exp(x) * grid.rho                               #  $G'(z)$ 
    G = reverse_cumulative(-dG, dx=grid.dz) \
        + np.exp(x[-1] - grid.z_max) / (1.0 - lam_p)        #
        right tail
    return Slice(x=x, dx=dx, G=G, dG=dG, m=m,
        lam=(lam_m, lam_p), theta=theta)

```

Both curves carry exact nodal derivatives, so off-grid reads are cubic Hermite — fourth-order, no spline solve, nodal values reproduced exactly at nodes.

A.2 Call evaluation

Listing 2: Pricing one strike: two ledger entries, the cash leg in log space, and the beyond-grid asymptote.

```

def call_price(slc, k, grid):
    if k > slc.x[-1]:                                       # beyond
        grid, right
        z_r = grid.z_max + (k - slc.x[-1]) / slc.lam[1]   # eq.
        (beyondgrid)
        return np.exp(k - z_r) * slc.lam[1] / (1.0 - slc.lam[1])
    if k < slc.x[0]:                                       # beyond
        grid, left:
        return 1.0 - np.exp(k) + call_asymptote_left(slc, k, grid)
        # parity
    z_k = hermite_root(slc.x, slc.dx, k)                   # Newton-
        polished
    G_k = hermite_eval(slc.G, slc.dG, z_k)
    cash = np.exp(k - np.logaddexp(0.0, z_k))              #  $e^k (1-
        u_k)$ , stable
    return G_k - cash                                       # eq. (
        call)

```

The one line a porter will be tempted to simplify is the `cash` line. Do not: `exp(k)*(1-expit(z))` is exactly the rank-saturation bug of section 12, invisible until a near-wall slice is priced at a far strike, then worth 1.9×10^{-2} of forward in phantom butterfly.

A.3 The analytic sensitivity pass

Listing 3: All Jacobian columns in one pass over the build’s arrays; eqs. (pass1)–(pass3) and (varswaprow).

```
def sensitivities(slc, grid):
    # basis[j] = d g / d theta_j on the grid: (1-u), u, P_n(1-2u)
    B = np.vstack([1 - grid.u, grid.u, grid.Pmat])          # (P,
    # n_grid)
    dxbar = cumulative_simpson(slc.dx * B, dx=grid.dz,
                                anchor=grid.mid, axis=1)    # pass 1
    w = np.exp(slc.x) * grid.rho                             # e^x rho
    # reused
    dm = -(w @ dxbar.T) * grid.dz_weights_total            # pass 2
    # (dot)
    dQ = dm[:, None] + dxbar                                # d x / d
    # theta
    integ = w[None, :] * dQ                                 # d(e^x
    # rho)/d theta
    dG = reverse_cumulative(integ, dx=grid.dz, axis=1)      # pass 3
    dvs = -2.0 * trapz(dQ * grid.rho, dx=grid.dz, axis=1)  # var-
    # swap row
    return dG, -integ, dvs                                  # nodal values, nodal slopes, w_vs
    # row

def dprice_dtheta(slc, sens, k):
    z_k = hermite_root(slc.x, slc.dx, k)
    return hermite_eval(sens.dG, sens.dGslope, z_k)         # thm (
    # envelope)
```

By theorem 10.1 the strike root is *not* differentiated — the columns are read off the nodal ledger sensitivities at a frozen root. Tail-correction terms of (54) contribute their own derivatives through $\bar{x}(\pm Z)$ and $\partial \lambda_{\pm} / \partial \theta = \lambda_{\pm} \cdot (1, 0, 1, 1, \dots) / \lambda_{\pm} \cdot (0, 1, 1, -1, \dots)$; they are two extra scalar rows per column and are elided above for space, not optional in practice.

A.4 The monotonicity certificate

Listing 4: Fritsch–Carlson margin for a Hermite interpolant with nodal derivatives `d` and values `y`.

```
def monotone_margin(y, d, dz, flat_tol=1e-9):
    s = np.diff(y) / dz                                     # secant slopes
    d0, d1 = d[:-1], d[1:]
    active = ~((np.abs(s) < flat_tol) & (np.abs(d0) < flat_tol)
               & (np.abs(d1) < flat_tol))                  # underflowed-flat
    # segments
    m = np.minimum.reduce([d0, d1, 3*s - d0, 3*s - d1])
    return m[active].min()                                  # certified iff > 0
```

Increasing curves are certified directly, decreasing ones (the ledger) by negation; segments below `flat_tol` are certified flat-to-tolerance, a stated caveat of the certificate.

A.5 Defaults

Quantity	Default	Quantity	Default
Basis order N	16 (cap 24)	Order guard	(45), floor $\min(N, 6)$
Grid half-width $Z_{\max}$	40	Grid points	8001 (opt. 2001)
Wall guard	$\lambda_+ \leq 1 - 10^{-6}$	Overflow budget	$\max g, \max \bar{x} \leq 700$
Ridge α , power ν	$10^{-6}, 1$ (rows $n \geq 4$)	Vega floor η	10^{-4}
Barrier	$\log(1 + e^{50(\lambda_+ - 0.90)})$	Haircut h	0.005 (vol)
Solver	trust-region LSQ	Tolerances	10^{-10} (x/f/g), max 4000 evals
Chart	logistic ($\lambda_+ = \text{expt } \xi$)	Calendar nodes	$\max(49, 4N + 1)$, $\cos^2$ taper
Belly certificate	801 pts, $g_D \geq -10^{-4}$	Cold start	$s = \sqrt{3}w_0/\pi$

Table 4: Reference-implementation defaults. Solver tolerances sit roughly six orders below the typical vol-bp fit budget: tight enough to be irrelevant to quality, loose enough to stop the trust region polishing noise.

A.6 Practical recommendations

1. Port the constant-speed audit first (section 4.3); it catches grid, anchoring, normalization, and interpolation errors in one closed-form comparison.
2. Keep the two cumulative curves and their nodal derivatives together as one object; every consumer (pricing, Jacobian, density, calendar, certificates) reads that object and nothing else.
3. Never expose raw (L, R) as user-facing tail controls; expose $(\log \lambda_-, \log \lambda_+)$ through the endpoint chart, or the market-package layer if a desk vocabulary is wanted. The reference implementation additionally provides a local chart whose leading coordinates are the exact ATM handles (built by right-inverting the handle Jacobian, with the remaining directions spanning its kernel) and delta-method uncertainty bands priced from the same sensitivity pass; both are conveniences over this paper’s objects, not new theory.
4. Run the certificates in production, not in development: the Fritsch–Carlson margin and belly certificate are microseconds to milliseconds, and the day they fail is the day they pay for themselves.
5. When the audit and a theorem disagree, believe the audit and find the floating-point failure; section 12’s catalogue is the complete list of the ones found so far.

B Deferred proofs

B.1 Proof of proposition 8.1

The smile is defined implicitly by $B(k, w(k)) = c(k)$. The plan: compute the five ATM partials of B , differentiate the defining identity twice, solve, and convert to volatility units. Throughout, $w_0 = w(0)$, $d = \sqrt{w_0}/2$, and all partials are evaluated at $(k, w) = (0, w_0)$.

Step 1: the five Black partials at the money. From (3), with $d_{\pm} = -k/\sqrt{w} \pm \sqrt{w}/2$, the identity $e^k \varphi(d_-) = \varphi(d_+)$ (verified by $d_+^2 - d_-^2 = (d_+ + d_-)(d_+ - d_-) = (-2k/\sqrt{w}) \cdot \sqrt{w} = -2k$) gives the standard cancellations:

$$\partial_k B = \varphi(d_+) \cdot \left(-\frac{1}{\sqrt{w}}\right) - e^k \Phi(d_-) - e^k \varphi(d_-) \cdot \left(-\frac{1}{\sqrt{w}}\right) = -e^k \Phi(d_-),$$

$$\partial_w B = \varphi(d_+) \partial_w d_+ - e^k \varphi(d_-) \partial_w d_- = \varphi(d_+) \partial_w (d_+ - d_-) = \frac{\varphi(d_+)}{2\sqrt{w}}.$$

Differentiating these once more, using $\varphi'(t) = -t \varphi(t)$:

$$\begin{aligned}\partial_{kk} B &= -e^k \Phi(d_-) - e^k \varphi(d_-) \partial_k d_- = -e^k \Phi(d_-) + \frac{\varphi(d_+)}{\sqrt{w}}, \\ \partial_{kw} B &= \partial_k \left[\frac{\varphi(d_+)}{2\sqrt{w}} \right] = \frac{-d_+ \varphi(d_+)}{2\sqrt{w}} \cdot \left(-\frac{1}{\sqrt{w}} \right) = \frac{d_+ \varphi(d_+)}{2w}, \\ \partial_{ww} B &= \frac{-d_+ \varphi(d_+) \partial_w d_+}{2\sqrt{w}} - \frac{\varphi(d_+)}{4w^{3/2}}, \quad \partial_w d_+ = \frac{k}{2w^{3/2}} + \frac{1}{4\sqrt{w}}.\end{aligned}$$

At $k = 0$: $d_+ = d$, $d_- = -d$, $\Phi(d_-) = 1 - \Phi(d)$, $\partial_w d_+ = 1/(4\sqrt{w_0})$, so

$$\begin{aligned}\partial_k B &= -(1 - \Phi(d)), \quad \partial_w B = \frac{\varphi(d)}{2\sqrt{w_0}}, \quad \partial_{kk} B = -(1 - \Phi(d)) + \frac{\varphi(d)}{\sqrt{w_0}}, \\ \partial_{kw} B &= \frac{\varphi(d)}{4\sqrt{w_0}}, \quad \partial_{ww} B = -\frac{\varphi(d)}{4w_0^{3/2}} - \frac{\varphi(d)}{16\sqrt{w_0}}.\end{aligned}\tag{58}$$

Step 2: first derivative. Differentiating $B(k, w(k)) = c(k)$ once and solving,

$$w'(0) = \frac{c'(0) - \partial_k B}{\partial_w B} = \frac{-(1 - u_*) + (1 - \Phi(d))}{\varphi(d)/(2\sqrt{w_0})} = \frac{2\sqrt{w_0}}{\varphi(d)} (u_* - \Phi(d)) = \frac{2\sqrt{w_0}}{\varphi(d)} \Delta,$$

which is (38).

Step 3: second derivative. Differentiating twice,

$$\partial_{kk} B + 2 \partial_{kw} B w'(0) + \partial_{ww} B w'(0)^2 + \partial_w B w''(0) = c''(0),$$

and solving for $w''(0)$ with $c''(0) = f_* - (1 - u_*)$ and (58):

$$\begin{aligned}w''(0) &= \frac{2\sqrt{w_0}}{\varphi(d)} \left[f_* - (1 - u_*) + (1 - \Phi(d)) - \frac{\varphi(d)}{\sqrt{w_0}} - \frac{\varphi(d)}{2\sqrt{w_0}} w'(0) + \left(\frac{\varphi(d)}{4w_0^{3/2}} + \frac{\varphi(d)}{16\sqrt{w_0}} \right) w'(0)^2 \right] \\ &= \frac{2\sqrt{w_0}}{\varphi(d)} f_* + \underbrace{\frac{2\sqrt{w_0}}{\varphi(d)} \Delta - 2}_{= w'(0)} - w'(0) + \left(\frac{1}{2w_0} + \frac{1}{8} \right) w'(0)^2 = \frac{2\sqrt{w_0}}{\varphi(d)} f_* - 2 + \left(\frac{1}{2w_0} + \frac{1}{8} \right) w'(0)^2,\end{aligned}$$

the promised (39): the digital-mismatch terms cancel against the cross term exactly.

Step 4: volatility units. With $\sigma(k) = \sqrt{w(k)/\tau}$: $\sigma' = w'/(2\sqrt{\tau w})$ and $\sigma'' = w''/(2\sqrt{\tau w}) - w'^2/(4\sqrt{\tau} w^{3/2})$. At $k = 0$, substituting $w'(0) = 2\sqrt{w_0} \Delta / \varphi(d)$:

$$\sigma'(0) = \frac{\Delta}{\varphi(d)\sqrt{\tau}}, \quad \sigma''(0) = \frac{1}{\sqrt{\tau}} \left[\frac{f_*}{\varphi(d)} - \frac{1}{\sqrt{w_0}} + \left(\frac{1}{2w_0} + \frac{1}{8} \right) \frac{w'(0)^2}{2\sqrt{w_0}} - \frac{w'(0)^2}{4w_0^{3/2}} \right],$$

and the bracket's last two terms combine: $\left(\frac{1}{2w_0} + \frac{1}{8} \right) \frac{1}{2\sqrt{w_0}} - \frac{1}{4w_0^{3/2}} = \frac{1}{4w_0^{3/2}} + \frac{1}{16\sqrt{w_0}} - \frac{1}{4w_0^{3/2}} = \frac{1}{16\sqrt{w_0}}$, giving with $w'(0)^2 = 4w_0 \Delta^2 / \varphi(d)^2$ the final form

$$\sigma''(0) = \frac{1}{\sqrt{\tau}} \left[\frac{f_*}{\varphi(d)} - \frac{1}{\sqrt{w_0}} + \frac{\sqrt{w_0}}{4} \left(\frac{\Delta}{\varphi(d)} \right)^2 \right],$$

which is (40). □

B.2 Proof of proposition 5.3

Setup. Let X^* satisfy the hypothesis: quantile $Q^*(u) = x^*(\log u)$ with $x^*(z) = e^{g^*(\Lambda(z))}$ for a continuous $g^*: [0, 1] \rightarrow \mathbb{R}$ with $g^*(1) < 0$, and $\mathbb{E}[e^{X^*}] = 1$. Write $\lambda_\pm^*$ for its endpoint scales and fix $\bar{\lambda} \in (\lambda_+^*, 1)$.

Step 1: polynomial approximation inside the family. By the Weierstrass approximation theorem, choose polynomials p_N of degree at most N with $\varepsilon_N := \|p_N - g^*\|_\infty \rightarrow 0$. Every polynomial of degree $\leq N$ is exactly representable in the basis of (13) (the endpoint chord spans degrees 0, 1 and $P_2, \dots, P_N$ the rest), so p_N defines a candidate slice with coefficients θ_N . Admissibility: its right endpoint value is $p_N(1) \leq g^*(1) + \varepsilon_N < \log \bar{\lambda} < 0$ for all N past some N_0 — admissibility is an open condition, so uniform approximation lands inside it. Fix such N hereafter; write $g_N = p_N$, and let $\bar{x}_N, m_N, x_N, Q_N, c_N$ be the associated objects.

Step 2: transports converge, with a uniform tail envelope. Pointwise, $|e^{g_N} - e^{g^*}| \leq e^{g^*}(e^{\varepsilon_N} - 1)$, so for every z ,

$$|\bar{x}_N(z) - \bar{x}^*(z)| \leq (e^{\varepsilon_N} - 1) \int_0^{|z|} e^{g^*(\Lambda(\pm t))} dt \leq (e^{\varepsilon_N} - 1) e^{\|g^*\|_\infty} |z|.$$

For the envelope, no derivative information about g_N is needed — sup-norm closeness suffices. Since g^* is continuous with $g^*(1) = \log \lambda_+^* < \log \bar{\lambda}$, pick z_0 and $\delta > 0$ with $g^*(\Lambda(t)) \leq \log \bar{\lambda} - \delta$ for all $t \geq z_0$; then for all N with $\varepsilon_N < \delta$, $e^{g_N(\Lambda(t))} \leq \bar{\lambda}$ on $[z_0, \infty)$, so integrating from z_0 ,

$$e^{\bar{x}_N(z)} \leq C_1 e^{\bar{\lambda}z} \quad (z \geq 0), \quad e^{\bar{x}_N(z)} \leq C_1 \quad (z \leq 0), \quad (59)$$

with $C_1 = \exp(z_0 e^{\|g^*\|_\infty + \delta})$ independent of N (the $z \leq 0$ bound holds because $\bar{x}_N$ is increasing with $\bar{x}_N(0) = 0$).

Step 3: normalizers and shifts converge. $I_N = \int e^{\bar{x}_N} \rho dz$. By Step 2 the integrands converge pointwise to $e^{\bar{x}^*} \rho$ and are dominated by $C_1 e^{\bar{\lambda}z} \rho(z) \leq C_1 e^{(\bar{\lambda}-1)z}$ on the right and $C_1 e^z$ on the left, both integrable; dominated convergence gives $I_N \rightarrow I^*$ and $m_N = -\log I_N \rightarrow m^*$. Hence $x_N \rightarrow x^*$ uniformly on every compact z -interval, i.e. $Q_N \rightarrow Q^*$ uniformly on compact subsets of $(0, 1)$.

Step 4: calls converge uniformly. Couple all the laws by one uniform rank: $Y_N = e^{Q_N(U)}$, $Y^* = e^{Q^*(U)}$. For every k ,

$$|c_N(k) - c^*(k)| = |\mathbb{E}[(Y_N - e^k)^+ - (Y^* - e^k)^+]| \leq \mathbb{E}|Y_N - Y^*|,$$

since $t \mapsto (t - e^k)^+$ is 1-Lipschitz — one bound, uniform in k . Now $\mathbb{E}|Y_N - Y^*| = \int_0^1 |e^{Q_N} - e^{Q^*}| du \rightarrow 0$: the integrand tends to zero pointwise (Step 3) and is dominated by $e^{Q_N} + e^{Q^*} \leq 2C_2(1-u)^{-\bar{\lambda}} + 2C_2$, integrable because $\bar{\lambda} < 1$. Here C_2 — the only place uniform control of the shifts is needed — is $C_2 = C_1 e^{\sup_N |m_N|} \cdot 2^{\bar{\lambda}}$: the factor C_1 from (59), the shift factor finite because $m_N \rightarrow m^*$ (Step 3) so $\sup_N |m_N| < \infty$, and the last factor from translating the envelope through the two-sided bound $\frac{1}{2}e^{-z} \leq 1 - \Lambda(z) \leq e^{-z}$ for $z \geq 0$.

Step 5: Wasserstein convergence of the return laws. For laws on $\mathbb{R}$ with finite p -th moments, $p \geq 1$, the quantile coupling is optimal: $W_p(Y_N, Y^*)^p = \int_0^1 |e^{Q_N} - e^{Q^*}|^p du$ [21, Ch. 2]. For $1 \leq p < 1/\lambda_+^*$ (a nonempty range, since $\lambda_+^* < 1$) choose $\bar{\lambda} \in (\lambda_+^*, \min(1, 1/p))$ in Step 2; then the integrand is dominated by $2^p C_2^p ((1-u)^{-p\bar{\lambda}} + 1)$ with $p\bar{\lambda} < 1$, integrable, and pointwise convergence again finishes it. (For $p > 1/\lambda_+^*$ the target law itself lacks the moment, so no metric statement is attempted.)

Step 6: exclusions. At any finite order, x'_N is continuous and bounded between two positive constants on every compact z -set; hence Q_N is a strictly increasing homeomorphism of $(0, 1)$ onto an unbounded interval. An atom of the law is a jump of the quantile function; a zero-density gap is a flat of it; bounded support means Q bounded; default mass at zero means $Q = -\infty$ on a rank interval — each impossible for such Q_N . A Gaussian (or any super-exponential) upper tail forces $g(u) \rightarrow -\infty$ as $u \rightarrow 1$ (section 3), impossible for a polynomial.

For the closure dichotomy, let $g_N \rightarrow g_\infty$ uniformly on $[0, 1]$ with every g_N admissible; g_∞ is continuous and $g_\infty(1) = \lim_N g_N(1) \leq 0$.

Branch $g_\infty(1) < 0$. Steps 2–4 apply verbatim with g_∞ as the target: normalizers, shifts, and quantiles converge (locally uniformly), the call curves converge uniformly — pointwise quantile convergence gives weak convergence — and the limit law is of the hypothesis class: positive continuous density, exponential tails, admissible scales.

Branch $g_\infty(1) = 0$. Weak convergence can fail, and the paper’s own solvable slice shows it: take $g_N \equiv \log(1 - 1/N)$, constant speeds approaching the wall. By (19), $I_N = \pi s_N / \sin(\pi s_N)$ with $s_N = 1 - 1/N$, so $I_N = \pi(1 - 1/N) / \sin(\pi/N) \rightarrow \infty$, hence $m_N = -\log I_N \rightarrow -\infty$ and every quantile $Q_N(u) = m_N + s_N \logit u \rightarrow -\infty$: for any fixed level q , $\mathbb{P}(X_N \leq q) \rightarrow 1$, the mass escapes to $-\infty$, and no weak limit exists. Suppose instead that the laws *do* converge weakly to a probability law μ . First, $\{m_N\}$ is bounded: above, because $I_N \geq \int_{-1}^1 e^{\bar{x}_N} \rho dz$ is bounded away from zero by locally uniform convergence of $\bar{x}_N$; and below, because $m_{N_j} \rightarrow -\infty$ along a subsequence would give $Q_{N_j}(u) = m_{N_j} + \bar{x}_{N_j}(\logit u) \rightarrow -\infty$ at every rank, i.e. $\mathbb{P}(X_{N_j} \leq q) \rightarrow 1$ for every q , contradicting the tightness of a weakly convergent sequence. Next, any subsequential limit $\bar{m}$ of m_N gives $Q_N \rightarrow \bar{m} + \bar{x}_\infty \circ \logit$ pointwise on $(0, 1)$ along that subsequence, and pointwise quantile convergence (at continuity points — here everywhere) is equivalent to weak convergence on $\mathbb{R}$; so μ ’s quantile equals $\bar{m} + \bar{x}_\infty \circ \logit$, which pins $\bar{m}$ and shows the whole sequence m_N converges. The limit quantile is C^1 in log-odds with derivative $e^{g_\infty(\Lambda(z))}$, bounded below by $e^{\min_{[0,1]} g_\infty} > 0$; hence μ has support all of $\mathbb{R}$ and, by the computation (14) with g_∞ in place of g , a strictly positive continuous density $u(1-u)e^{-g_\infty(u)}$ at rank u — no atom, no gap, no bounded support, and no Gaussian-or-faster tail (which would force $g_\infty(u) \rightarrow -\infty$ at an endpoint). Its right tail sits at the wall: for every $\delta > 0$ there is z_δ with $e^{g_\infty(\Lambda(z))} \geq 1 - \delta$ for $z \geq z_\delta$, so the quantile grows at asymptotic rate at least $1 - \delta$, and the moment-strip computation of proposition 6.2 (with $1 - \delta$ in place of λ_+ , δ arbitrary) gives $\mathbb{E}_\mu[e^{rX}] = \infty$ for every $r > 1$; by Fatou, $\mathbb{E}_\mu[e^X] \leq \liminf_N \mathbb{E}[e^{X_N}] = 1$, with strict inequality possible. The boundary limit, when it exists, is thus not a member of the class of (i) — its tail scale has been lost — but neither is it any excluded law. In both branches, no excluded law is reached. $\square$

Remark B.1. Step 4’s one-line Lipschitz bound is the reason the paper prices through the rank coupling everywhere: the same trick gives the stability of every price-linear functional under parameter perturbations, and it is the continuous heart of the delta-method bands mentioned in appendix A.

C Data and reproducibility

C.1 The frozen snapshot

Every real-market number and figure in the paper derives from one frozen JSON artifact, captured once and never refetched:

- *Source:* a delayed consolidated US listed-options feed; after-hours book of the Monday 2026-08-03 US session (reference date 2026-08-03, capture timestamp 2026-08-04 02:08 UTC).

- *Universe*: SPY (spot 757.67) and NVDA (spot 206.80), eight expiries each, 2026-08-05 through 2027-12-17 — maturities from 2 calendar days to 1.3726 years. 16 fitted nodes in total.
- *Fit recipe* (stamped in the file’s manifest): LQD at order $N = 16$ under the guard (45), logistic chart, haircut target (44) with $h = 0.005$, equal quote weights, ridge $\alpha = 10^{-6}$, $\nu = 1$, calendar rows (53) active. No per-node tuning of any kind.
- *Quality*: all lit nodes fitted and ready; median quote-fit rms 5.9 vol bp, worst 24.6 vol bp (the shortest NVDA node).
- *Self-containment*: the file embeds the normalized chains, per-node prepared quotes with forward and discount provenance, fitted parameters, and fitted curves — everything needed to re-run any fit or re-price any figure offline, with no feed access.

One family-level advisory is disclosed rather than suppressed, because section 11 uses it as a live illustration: a full-grid *vol-space* calendar audit at equal log-moneyness (both legs read on the far node’s clock) flags a 1840.6 vol bp crossing between the two shortest SPY expiries — two and four calendar days from the reference date — at $k = +0.981$, far outside the pair’s common quote span $[-0.028, +0.016]$, where the call prices are 5.4×10^{-107} and 1.6×10^{-26} of forward (bounded by 10^{-26}). A price-space audit over all adjacent pairs on the full display grid finds a worst violation of 2.2×10^{-15} of forward — round-off. Per-node quality is clean; the crossing lives entirely in the extrapolated region that section 11.2 deliberately declines to constrain.

C.2 Synthetic benchmarks

Two synthetic targets appear in section 13. The *double-hump* target is a two-component lognormal mixture in the gross return, martingale-normalized, with component weights, centers, and widths fixed in the figure-generation script and printed in the figure captions’ companion macro file; it is quoted as noiseless implied volatilities on a realistic strike ladder, so it measures approximation capacity, not market fit. The *constant-speed* audit needs no target data at all: it compares the implementation against the closed forms of section 4.3.

C.3 Regeneration

All figures and every data-derived number in the text (as LaTeX macros) are produced by a single generator script that reads only the frozen snapshot and the synthetic specifications — never the network. Real nodes are rebuilt from their frozen coefficient vectors, not refitted, and the rebuild is gated: the rebuilt display curves reproduce the frozen ones to 0.000 vol bp worst-case across all 16 nodes, so every figure prices off exactly the published fits. Regenerating the paper is: run the generator (one pass, deterministic, fixed seeds for the audit battery), then compile. Timing tables are the exception to determinism in value but not in protocol: they rerun the interleaved median/IQR measurement of section 12 on the build machine, and are expected to differ across machines — the text quotes them only with that caveat. The snapshot file itself is the citable artifact: refetching from the feed would produce a different book and different (if statistically similar) numbers, and no claim in the paper depends on which day was frozen.

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